

01 — Project Overview & Architecture

1. What Is This Project?

The **CAG Website** is a large-scale, multi-site **Content Management System (CMS)** built for the **Comptroller and Auditor General of India (CAG)**. It serves as the primary digital platform for publishing audit reports, circulars, notices, recruitment information, training data, and administrative content across **multiple regional sub-sites** (state audit offices, ICISA training institute, NAAA academy, etc.).

2. Technology Stack

Layer	Technology	Version
Language	PHP	8.3
Framework	CakePHP	5.3.6
Database	PostgreSQL	16+
Web Server	Apache	2.4 (mod_rewrite)
Containerisation	Docker	Multi-stage build
OS (Container)	Debian Bookworm	12.x
PDF Generation	mPDF	8.2
Excel Generation	PhpSpreadsheet	2.1
QR Codes	endroid/qr-code	4.8
Authentication	cakephp/authentication	4.2

3. High-Level Architecture Diagram

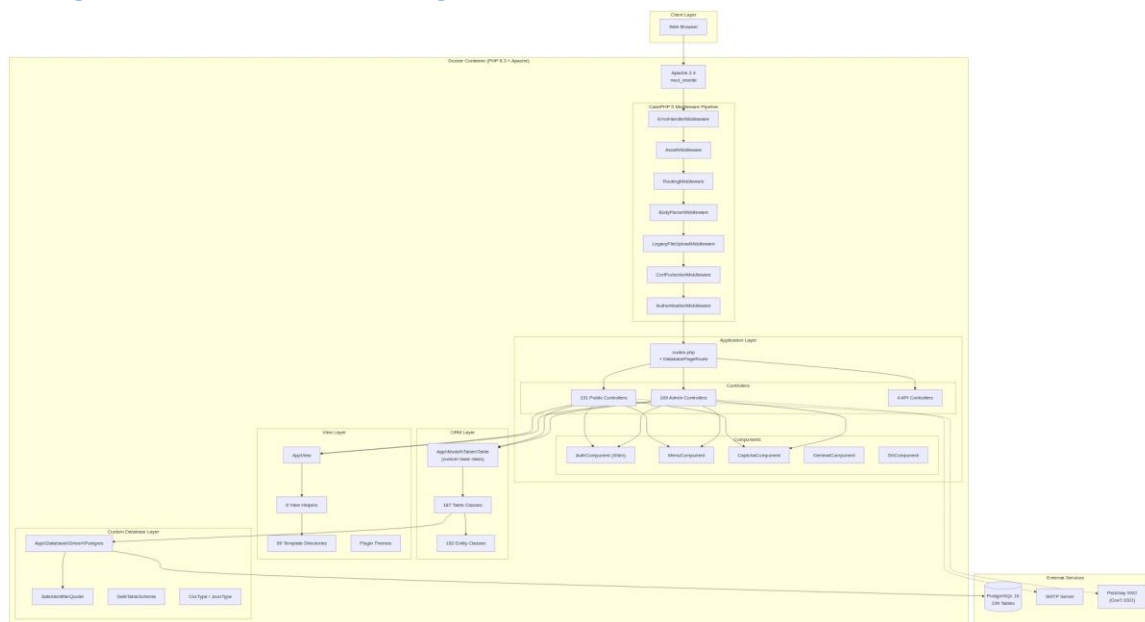


Figure: 01_project_overview Diagram 1

4. Project Directory Structure

```
CAG_Website/
├── config/                                # Application configuration
│   └── bootstrap.php                    # Bootstrap: class aliases, JCrypton stub,
error traps
│   ├── routes.php                      # All route definitions (454 lines)
│   ├── constants.default.php           # Application constants (role IDs, paths, etc.)
│   └── app.default.php                 # CakePHP core config (DB, cache, sessions,
email)
│   └── .env.default                    # Environment variable templates
├── src/                                # Application source code
│   ├── Application.php                 # Bootstrap, middleware queue, auth service
│   ├── Controller/
│   │   ├── AppController.php           # Public base controller (667 lines)
│   │   ├── Admin/
│   │   │   ├── AppController.php       # Admin base controller (935 lines)
│   │   │   ├── UsersController.php     # Login/logout/user management (2131 lines)
│   │   │   └── ... (169 files)
│   │   ├── Api/                       # REST API controllers
│   │   ├── Component/                 # 8 shared components
│   │   ├── Migration/                 # Data migration utilities
│   │   └── ... (131 public controllers)
│   ├── Model/
│   │   ├── Table/
│   │   │   ├── Table.php              # Custom base Table class (26KB)
│   │   │   └── ... (187 table classes)
│   │   ├── Entity/                   # 182 entity classes
│   │   └── Validation/                # Custom file validation rules
│   ├── View/
│   │   ├── AppView.php               # SafePaginator + URL filters
│   │   ├── Helper/                   # 8 view helpers
│   │   └── Cell/                     # View cells
│   ├── Middleware/                   # 8 custom middleware classes
│   ├── Database/                     # Custom Postgres driver, quoter, types
│   ├── Routing/Route/                 # DatabasePageRoute (0(1) page routing)
│   ├── Filesystem/                   # File/Folder shims (CakePHP 3 compat)
│   └── Utility/                       # CsvSanitizerTrait
├── templates/                          # View templates (.php)
│   ├── Admin/                        # Admin panel templates
│   ├── Home/                         # Homepage templates
│   ├── element/                       # Reusable partial templates
│   ├── layout/                        # Layout wrappers (default, admin, home)
│   ├── error/                         # Error pages (400, 404, 500)
│   └── ... (89 template directories)
├── plugins/Themes/                    # Pluggable visual themes per sub-site
├── webroot/                           # Publicly accessible files (CSS, JS, uploads)
├── docker-compose.yml                  # Base Docker Compose (production)
├── docker-compose.prod.yml             # Production with local PostgreSQL
├── docker-compose.override.yml         # Dev overrides (volume mounts, Xdebug)
├── Dockerfile                          # Multi-stage PHP 8.3 + Apache image
├── entrypoint.sh                       # Container startup script
└── composer.json                       # PHP dependencies
```

5. Multi-Site Architecture

The CAG Website is a **single codebase** serving **multiple independent sub-sites**. Each sub-site has its own:

- URL prefix (e.g., `/cen/new-delhi-iv`, `/rly/mumbai-i`, `/naaa/naaa75years`)
- Theme (visual skin loaded as a CakePHP plugin from `plugins/Themes/`)
- Menu structure (configured via the `menus`, `menu_regions` tables)
- Content access rules (managed via `content_accesses` junction table)

The `websites` table defines all registered sub-sites. The `Websites::websiteCache()` method caches these into the `silver-website` cache for O(1) route resolution.

6. Key Design Patterns

Pattern	Implementation	Purpose
Compatibility Shim	<code>AuthComponent</code> , <code>FormHelper</code> , <code>Filesystem\File</code>	Bridges CakePHP 3 API on CakePHP 5 runtime
Custom Base Class	<code>App\Model\Table\Table</code>	Centralized SQL query rewriting for PostgreSQL
Database Page Route	<code>DatabasePageRoute</code>	O(1) CMS page resolution from cache
Middleware Pipeline	8 custom + 4 framework middleware	Security, auth, sanitization, i18n
Plugin Themes	<code>plugins/Themes/{ThemeName}/</code>	Per-sub-site visual customization
Role-Based Access	<code>RolesPermissions</code> + <code>ModulesAccesses</code> tables	Controls admin panel visibility per role

7. Scale Metrics

Metric	Count
Total PHP files in <code>src/</code>	500+
Database tables	199
Public page controllers	131
Admin CRUD controllers	169
Registered CakePHP routes	~300 (dynamic)
Lines of code (largest file)	<code>OrganisationChartController.php</code> — 136,450 bytes
Template directories	89

02 — Deployment Infrastructure & Production Environment

1. Docker Architecture Overview

The CAG Website runs as a **single containerized service** connecting to an **external PostgreSQL 16** database. The containerisation uses a **multi-stage Docker build** to keep the production image lean.

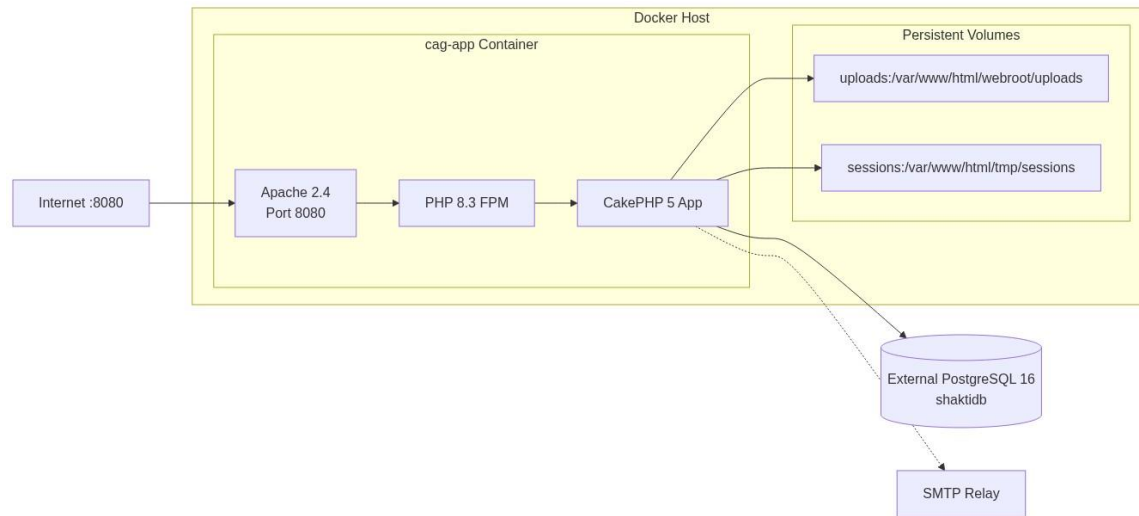


Figure: 02_deployment_infrastructure Diagram 1

2. Docker Compose Files

2.1 Base Compose — [docker-compose.yml](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/docker-compose.yml)

The base configuration for **any** environment:

Setting	Value	Purpose
Image	r-cag/cag-website:8.3-cake5	Pre-built or locally built image
Port	8080:8080	Apache listens on 8080 inside container
Timezone	Asia/Kolkata	IST for all date operations
Memory Limit	1 GB	Container memory cap
CPU Limit	1.5 cores	CPU allocation
Healthcheck	PHP fsockopen to port 8080 every 30s	Auto-restart on failure
Volumes	uploads + sessions (named Docker volumes)	Persists uploaded files and user sessions

Environment variables passed in:

- `SECURITY_SALT` — CakePHP encryption salt
- `DB_HOST`, `DB_PORT`, `DB_NAME`, `DB_USER`, `DB_PASSWORD`, `DB_SCHEMA` — PostgreSQL connection
- `SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `SMTP_TLS` — Email relay

2.2 Production Compose — [docker-compose.prod.yml](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/docker-compose.prod.yml)

Adds a **local PostgreSQL 17 container** for self-hosted deployments:

Service	Image	Configuration
cag-db	postgres:17	User: <code>shaktidb_admin</code> , DB: <code>shaktidb</code> , Port: <code>5433:5432</code>
cag-app	r-cag/cag-website:8.3-cake5	Depends on <code>cag-db</code> (health check), Schema: <code>public</code>

The app waits for the database healthcheck (`pg_isready`) before starting.

2.3 Dev Override — [docker-compose.override.yml](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/docker-compose.override.yml)

Automatically merged by Docker Compose during development:

Override	Effect
<code>APP_ENV=development</code>	Enables debug mode
<code>DEBUG=true</code>	Shows CakePHP error details
Volume <code>./var/www/html</code>	Live-mounts source code for hot reload
Volume <code>vendor_dev</code>	Separate named volume for <code>vendor/</code> to avoid shadow
<code>docker/php/dev.ini</code>	Dev PHP overrides (higher error reporting, Xdebug)
<code>host.docker.internal</code>	Allows Xdebug to reach IDE on host machine

3. Dockerfile — Multi-Stage Build

[Dockerfile](#) uses a **two-stage build**:

Stage 1: `vendor` (Composer Dependencies)

```
FROM composer:2 AS vendor
→ Copies composer.json + composer.lock
→ Runs `composer install --no-dev --optimize-autoloader`
→ Produces /app/vendor/ with all PHP dependencies
```

Stage 2: `runtime` (PHP 8.3 + Apache)

```
FROM php:8.3-apache AS runtime
```

System libraries installed:

- `libc-dev` — Intl extension
- `libpng-dev, libjpeg, libfreetype` — GD (image processing for captchas)
- `libpq-dev` — PostgreSQL PDO driver
- `libzip-dev` — ZIP handling
- `libonig-dev` — mbstring
- `libldap2-dev` — LDAP authentication

PHP extensions compiled:

`pdo_pgsql, pgsql, intl, gd, zip, mbstring, opcache, bcmath, exif, xml, ldap`

Apache modules enabled:

`rewrite, headers, expires`

Build-Time Operations

1. **Copy Apache config** from `docker/apache/cag.conf`
2. **Copy PHP configs** — `php.ini` and `opcache.ini`
3. **Create writable directories:** `tmp/cache/*, tmp/sessions, logs, webroot/uploads`

4. **Generate case-insensitive symlinks** via `ensure_case_symlinks()` function:

- Scans `templates/` and `plugins/Themes/` for directories
- Creates lowercase ↔ capitalized symlinks (e.g., `element` → `Element`)
- Required because CakePHP 5 expects `Element/`, `Layout/` etc. but Git preserves lowercase

5. **Remove sensitive configs:** Deletes `config/app_local.php` and `config/.env`

Final Image Labels

```
org.opencontainers.image.title = "CAG Website"
org.opencontainers.image.version = "8.3-cake5"
org.opencontainers.image.vendor = "r-cag"
```

4. Entrypoint Script

[entrypoint.sh](#) — Container startup:

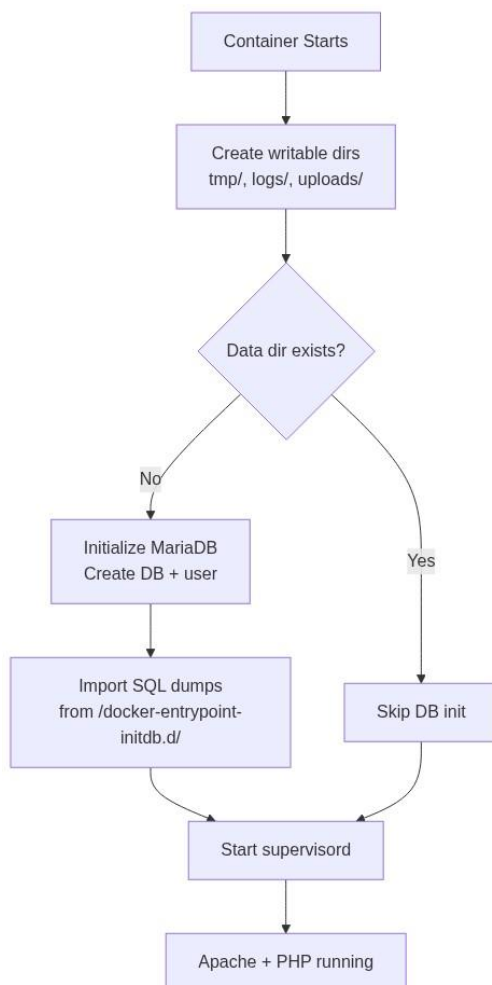


Figure: 02_deployment_infrastructure Diagram 2

> **Note:** The `entrypoint.sh` in the project root is a **legacy script** that references MariaDB. The actual production deployment uses `docker/entrypoint.sh` and connects to **external PostgreSQL**. The MariaDB references are vestigial from the original development environment.

5. Apache Configuration

`cag.conf`:

- VirtualHost on `*:8080`
- DocumentRoot: `/var/www/html/webroot`
- `AllowOverride All` — enables `.htaccess` for CakePHP URL rewriting
- `mod_rewrite` rules redirect all non-file/non-directory requests to `index.php`

6. PHP Configuration

`[php.ini](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/php.ini)`

Setting	Value	Purpose
<code>memory_limit</code>	<code>256M</code>	Maximum memory per request
<code>upload_max_filesize</code>	<code>100M</code>	Large document uploads
<code>post_max_size</code>	<code>100M</code>	Matches upload limit
<code>max_execution_time</code>	<code>300</code>	5-minute timeout for reports
<code>display_errors</code>	<code>Off</code>	Security: hide errors in production
<code>log_errors</code>	<code>On</code>	Errors written to log files

`[opcache.ini](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/opcache.ini)`

Setting	Value	Purpose
<code>opcache.enable</code>	<code>1</code>	Bytecode caching enabled
<code>opcache.memory_consumption</code>	<code>128</code>	128MB for opcode cache
<code>opcache.max_accelerated_files</code>	<code>10000</code>	Handles 500+ PHP files
<code>opcache.validate_timestamps</code>	<code>0</code>	Production only: no file stat checks
<code>opcache.revalidate_freq</code>	<code>0</code>	Immediate in dev (when <code>validate=1</code>)

7. Supervisor Configuration

`supervisord.conf`:

Manages the following services inside the container:

- **Apache** (`apache2ctl -D FOREGROUND`) — Main web server
- Optionally MariaDB (legacy, not used in PostgreSQL production)

8. Environment Variables Reference

Variable	Required	Default	Description
APP_ENV	Yes	production	production or development
DEBUG	Yes	false	Enable CakePHP debug mode
SECURITY_SALT	Yes	—	Encryption salt for sessions/cookies
DB_HOST	Yes	—	PostgreSQL hostname
DB_PORT	No	5432	PostgreSQL port
DB_NAME	Yes	—	Database name
DB_USER	Yes	—	Database username
DB_PASSWORD	Yes	—	Database password
DB_SCHEMA	No	cag	PostgreSQL schema name
TZ	No	Asia/Kolkata	Container timezone
SMTP_HOST	No	localhost	SMTP server for email
SMTP_PORT	No	25	SMTP port
SMTP_USER	No	—	SMTP authentication username
SMTP_PASSWORD	No	—	SMTP authentication password
SMTP_TLS	No	false	Enable TLS for SMTP

9. Deployment Commands

Development

```
# Start with live-mounted source code + debug
docker compose up -d --build
# → http://localhost:8080
```

Production (Self-Hosted PostgreSQL)

```
# Uses docker-compose.prod.yml with local Postgres container
docker compose -f docker-compose.prod.yml up -d
```

Production (External PostgreSQL)

```
# Copy and fill environment file
cp.env.docker.example.env
# Edit.env with real DB credentials
docker compose up -d
```

Build Image Only

```
docker build -t r-cag/cag-website:8.3-cake5 .
```

03 — Middleware Pipeline & Request Lifecycle

1. Complete Middleware Pipeline

Every HTTP request passes through this ordered pipeline before reaching a controller:

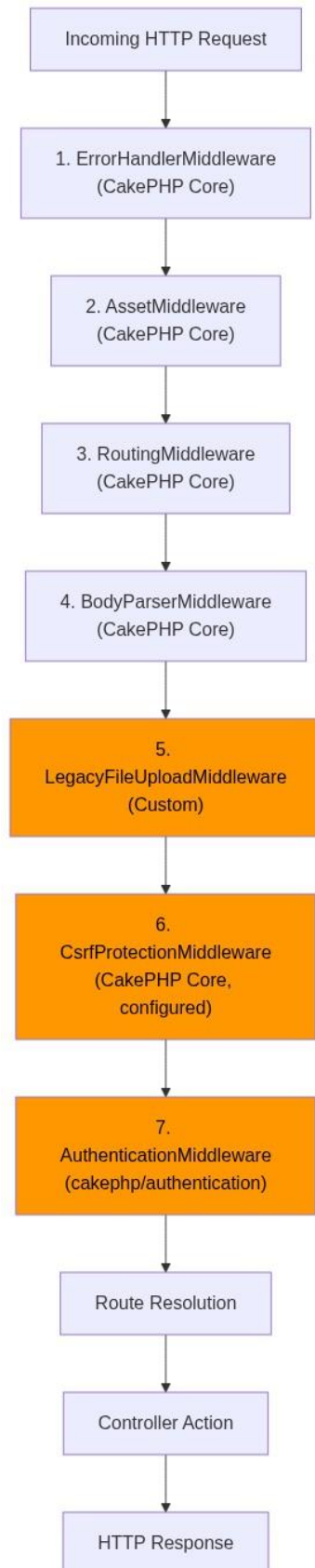


Figure: 03_middleware_pipeline Diagram 1

> **Note:** The middleware in [Application.php](#) is registered in the `middleware()` method. Additionally, `routes.php` registers **route-scoped middleware** (`LanguageRoutingMiddleware`) which runs only for matched routes.

2. Framework Middleware (Registered in Application.php)

2.1 ErrorHandlerMiddleware

- **Source:** CakePHP Core
- **Purpose:** Catches all uncaught exceptions and PHP errors, renders error pages
- **Configuration:** Uses `Configure::read('Error')` settings

2.2 AssetMiddleware

- **Source:** CakePHP Core
- **Purpose:** Serves plugin assets (CSS, JS, images) from plugin webroot directories
- **Configuration:** `cacheTime` from `Asset.cacheTime` config

2.3 RoutingMiddleware

- **Source:** CakePHP Core
- **Purpose:** Parses the URL and matches it against registered routes in `config/routes.php`
- **Produces:** Sets `controller`, `action`, `pass`, `prefix`, `plugin` params on the request

2.4 BodyParserMiddleware

- **Source:** CakePHP Core
- **Purpose:** Parses JSON and XML request bodies into `$request->getData()`

3. Custom Middleware (8 Files)

3.1 LegacyFileUploadMiddleware

- **File:** [LegacyFileUploadMiddleware.php](#)
- **Registered in:** `Application::middleware()` (global)
- **Purpose:** CakePHP 5 uses PSR-7 `UploadedFileInterface` objects for file uploads. Legacy controller code expects old-style `$_FILES` arrays (`name`, `tmp_name`, `error`, `size`, `type`). This middleware intercepts PSR-7 uploaded files and converts them back into nested arrays, merging them into the request's parsed body.
- **Impact:** Without this, every file upload form in the application would fail.

3.2 CsrfProtectionMiddleware (Configured)

- **File:** CakePHP Core, configured in [Application.php](#)

- **Configuration:**
 1. `httponly: true` — Cookie not accessible via JavaScript
 2. **Skipped for:** URLs containing `get-most-view-recent`, `login`, or action `getMostViewRecent`
 3. Only active when `PHP_SAPI !== 'cli'` (skipped during CLI/testing)

3.3 AuthenticationMiddleware

- **File:** `cakephp/authentication` plugin
- **Registered in:** `Application::middleware()` (global)
- **Purpose:** Runs the authentication service (`getAuthenticationService()`) to check session/form credentials
- **Details:** See [04_authentication_authorization.md](#)

3.4 LanguageRoutingMiddleware

- **File:** [LanguageRoutingMiddleware.php](#)
- **Registered in:** `routes.php` (route-scoped, not global)
- **Purpose:** Handles i18n URL routing. Reads the `{language}` route parameter (e.g., `en`, `hi`), sets `I18n::setLocale()`, and generates alternate-language links for the language switcher.
- **Redirect behavior:** If no language is specified in the URL, it redirects to the default language version (`/en/...`).

3.5 SanitizeRequestMiddleware

- **File:** [SanitizeRequestMiddleware.php](#)
- **Registered via:** Legacy CakePHP 3 `__invoke()` pattern (not PSR-15)
- **Purpose:** Security audit middleware that:

1. Checks all query parameters against a whitelist of ~60 allowed parameter names
2. Rejects unknown query parameters with a `NotFoundException` on public pages
3. Detects SQL injection patterns (`UNION`, `SELECT`, `DROP`, etc.) in query strings
4. Validates numeric-only parameters (`page`, `show_ut_only`)
5. Blocks requests with malicious patterns in `REQUEST_URI` and `QUERY_STRING`

3.6 BlockMaliciousUploadMiddleware

- **File:** [BlockMaliciousUploadMiddleware.php](#)
- **Purpose:** Blocks uploads of dangerous file types:
- **Blocked extensions:** `.php*`, `.phtml`, `.exe`, `.sh`, `.bat`, `.js`, `.html`, `.svg`, `.txt`, `.zip`
- **Blocked MIME types:** `application/x-php`, `text/x-php`, `application/x-executable`, `text/plain`, `application/zip`
- **Allowed extensions:** `.jpg`, `.jpeg`, `.png`, `.gif`, `.doc(x)`, `.xls(x)`, `.ppt(x)`, `.pdf`

3.7 FileValidationMiddleware

- **File:** [FileValidationMiddleware.php](#)
- **Purpose:** Deep file content validation on all POST requests:
- Validates file extensions against whitelist
- Validates MIME types using `finfo` (actual binary inspection)
- Only applies full validation on **front-end** (non-admin) POST requests
- Admin uploads have relaxed validation

3.8 HostHeaderMiddleware

- **File:** [HostHeaderMiddleware.php](#)
- **Purpose:** Prevents Host Header Injection attacks by comparing the incoming request's **Host** header against the configured **App.fullBaseUrl**. Throws **BadRequestException** on mismatch in production.

3.9 NoBackRequestMiddleware

- **File:** [NoBackRequestMiddleware.php](#)
- **Purpose:** Sets cache-control headers to prevent browser back-button from showing cached authenticated pages:

Cache-Control: no-store, no-cache, must-revalidate
Pragma: no-cache

3.10 CsrfCheckMiddleware

- **File:** [CsrfCheckMiddleware.php](#)
- **Purpose:** Additional CSRF token validation for specific routes/forms beyond the standard CakePHP middleware.

4. Request Lifecycle — End to End

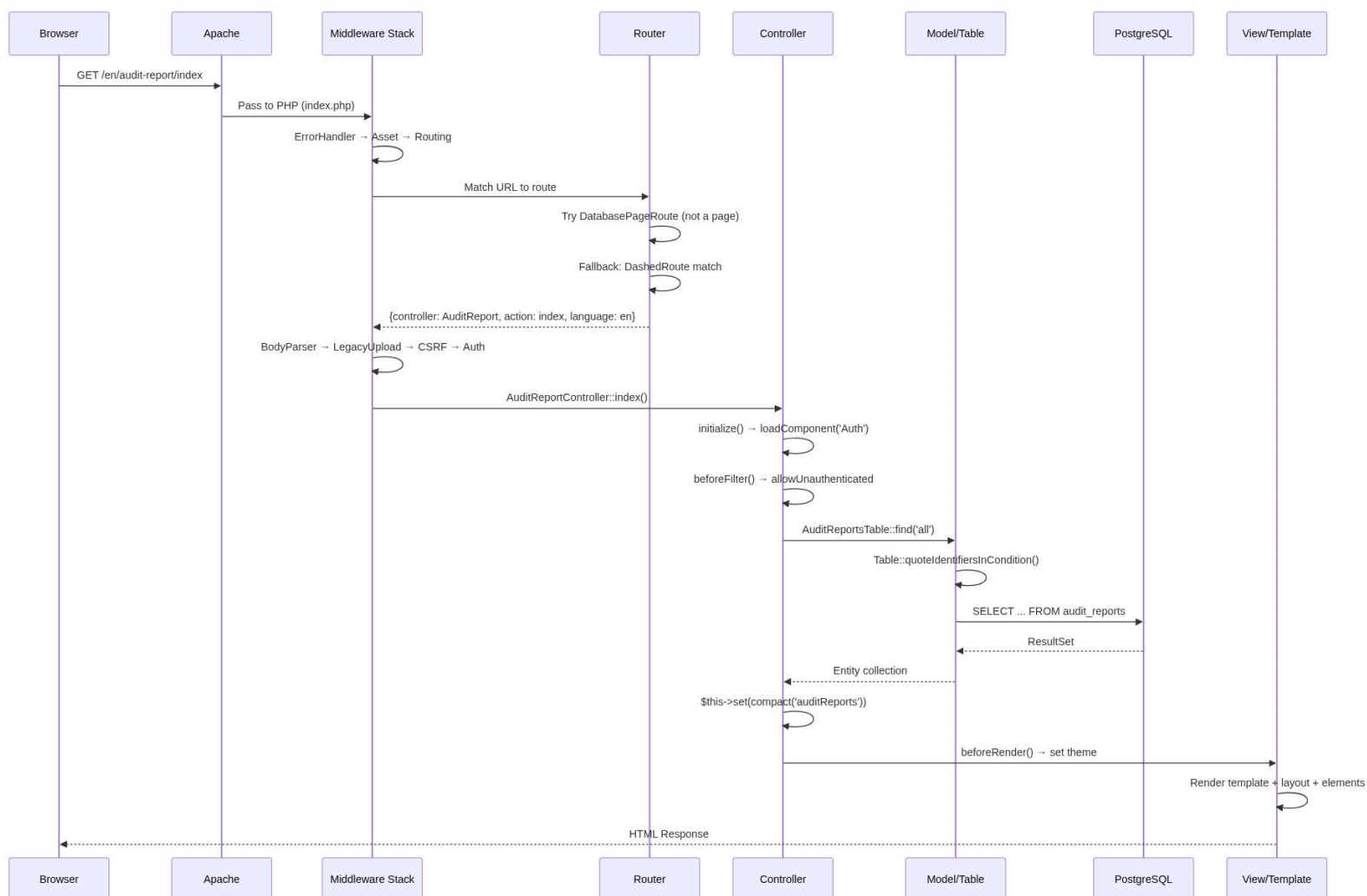


Figure: 03_middleware_pipeline Diagram 2

04 — Authentication & Authorization

1. Authentication Architecture

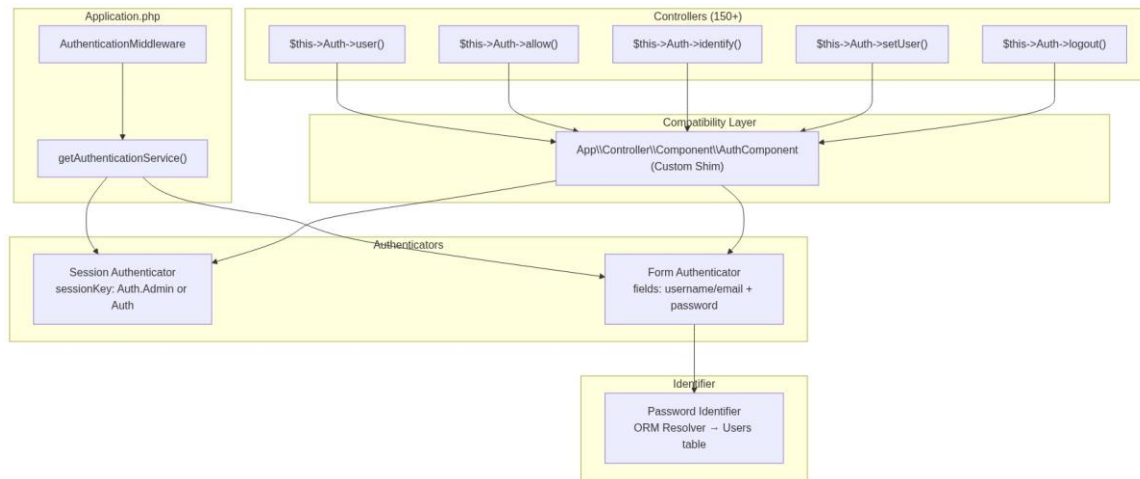


Figure: 04_authentication_authorization Diagram 1

2. Authentication Service Configuration

Defined in `Application::getAuthenticationService()`:

Session Authenticator

- **Session key:** `Auth.Admin` for admin-prefixed routes, `Auth` for public routes
- **Purpose:** Checks if a valid identity already exists in the PHP session

Form Authenticator

- **Login URLs accepted:**
- `/en/login`, `/hi/login`
- `/admin/en/users/login`, `/admin/hi/users/login`
- `/admin/users/login`, `/login`
- **Username fields:** Accepts both `username` and `email`
- **Password field:** `password`
- **Resolver:** CakePHP ORM → `Users` table
- **URL checker:** `MultiUrlChecker` (matches any of the listed login URLs)

Unauthenticated Redirect

All unauthenticated requests to admin routes redirect to:

```
/admin/users/login?redirect={original_url}
```

3. AuthComponent Compatibility Shim

File: [AuthComponent.php](#)

This is a **critical bridge** class. CakePHP 5 removed the core `AuthComponent`. Rather than rewriting 300+ controllers, the developers created a custom component with the same name that delegates to the new `Authentication` middleware:

Legacy Method	What It Does in the Shim
<code>\$this->Auth->user('field')</code>	Reads from session (<code>Auth.Admin.field</code> or <code>Auth.field</code>)
<code>\$this->Auth->allow(['action1', 'action2'])</code>	Calls <code>\$this->Authentication->allowUnauthenticated()</code>
<code>\$this->Auth->identify()</code>	Calls <code>\$this->Authentication->getResult()</code> , returns user array if successful
<code>\$this->Auth->setUser(\$user)</code>	Writes user data into the session under the appropriate key
<code>\$this->Auth->logout()</code>	Calls <code>\$this->Authentication->logout()</code> and clears session
<code>\$this->Auth->redirectUrl()</code>	Reads the <code>redirect</code> query parameter or returns login redirect config
<code>\$this->Auth->setConfig()</code>	Stores config locally (limited compatibility)

Session Key Scoping

- **Admin controllers:** Use `Auth.Admin` as the session key
- Set via `$this->Auth->sessionKey = 'Auth.Admin'` in [Admin/AppController.php](#)
- **Public controllers:** Use `Auth` as the session key
- This prevents admin sessions from leaking into public pages and vice versa.

4. Login Flow

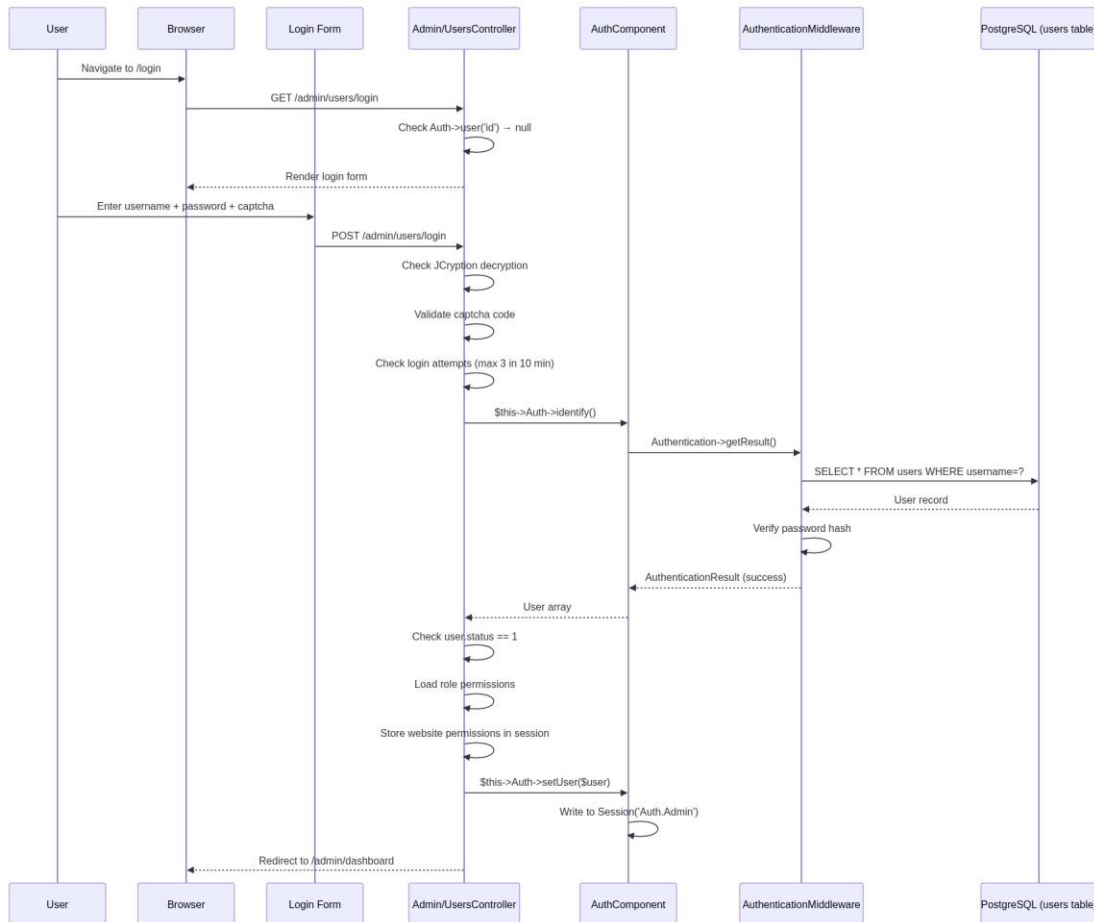


Figure: 04_authentication_authorization Diagram 2

Login Security Features

1. **Rate limiting:** Max 3 login attempts per username+IP within 10 minutes (PostgreSQL `INTERVAL '10 minutes'`)
2. **Captcha validation:** Required on login form
3. **JCryption:** Client-side password encryption (legacy, stubbed out)
4. **IP binding:** Admin session is bound to the client's IP address
5. **User-Agent binding:** Admin session validates the browser user-agent hasn't changed
6. **Concurrent login prevention:** Checks `login_token` and `login_token_at` to prevent simultaneous logins
7. **Token-based session validation:** Each admin request validates a session token

5. Authorization (Role-Based Access Control)

5.1 Role System

The `roles` table defines user roles. Key role IDs (from constants):

- Role 1: **Super Admin** — Full access
- Role 2: **Public User** — Redirected away from admin
- `KMS_ROLE_ID`: **KMS User** — Access to Knowledge Management
- `ICISA_MEMBER_ROLE_ID`: **ICISA Member** — Redirected to `/icisa`

5.2 Permission Tables

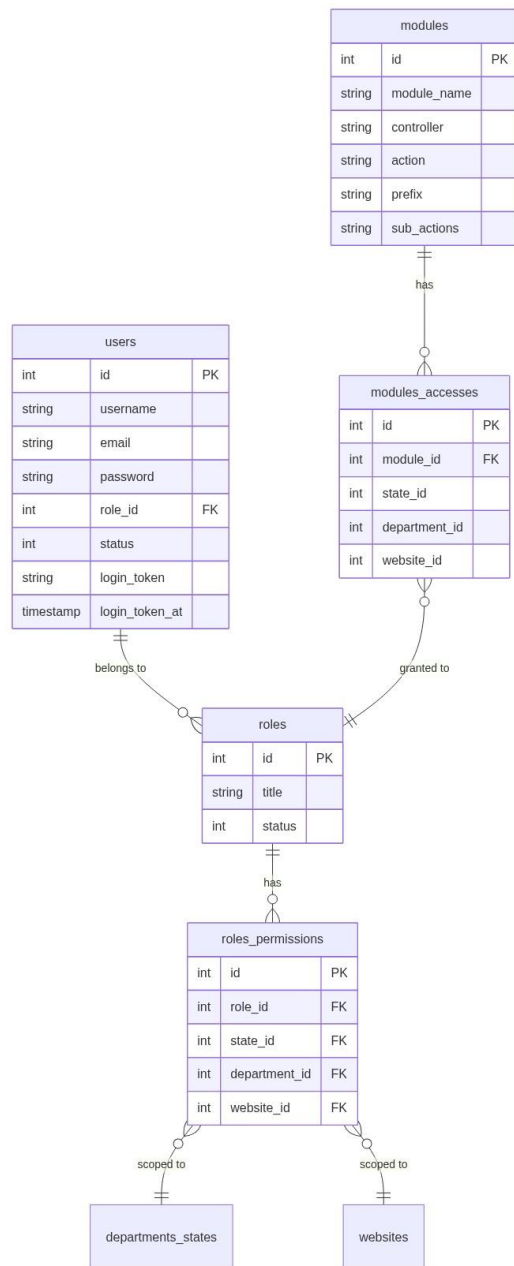


Figure: 04_authentication_authorization Diagram 3

5.3 How Admin Access Control Works

In [Admin/AppController::initialize\(\)](#):

1. Load `RolesPermissions` for the current user's `role_id`
2. Extract permitted `state_ids` and `department_ids`
3. Load `ModulesAccesses` — maps which modules (controllers) a role can access, filtered by `website_id`
4. The admin sidebar menu only shows modules the user has access to
5. Each admin controller can further restrict actions based on `$this->UserRolePermissions`

5.4 Public Page Access Control

In [AppController::initialize\(\)](#):

- All public controller actions are set to `allowUnauthenticated` in `beforeFilter()`
- Menu-level access is checked via `MenuComponent::hasAccess()` — if a menu region requires authentication but the user isn't logged in, they're redirected to `/login`

6. Password Management

Password Hashing

- Uses `Authentication\PasswordHasher\DefaultPasswordHasher` (bcrypt)
- Legacy code in `User` entity: `_setPassword()` auto-hashes on entity save

Password History

- `password_history` table stores previous password hashes
- Prevents password reuse

Password Reset Flow

- `forgotPassword` action sends reset email with token
- `resetPassword` action validates token and allows new password
- Token expiry is enforced

05 — Database ER Diagram (199 Tables)

The database contains **199 tables** in PostgreSQL 16. Below, they are organized into **functional domains** with focused ER diagrams for each domain.

1. Core CMS & Content Domain

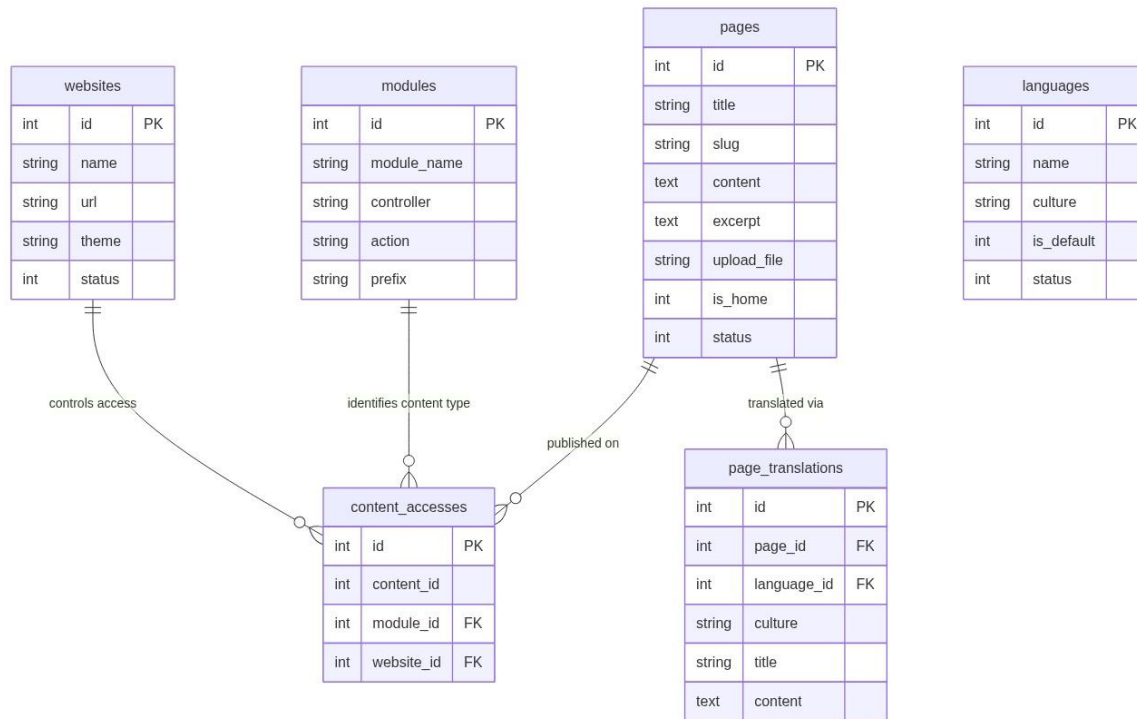


Figure: 05_database_er_diagram Diagram 1

2. User Management & Authentication Domain

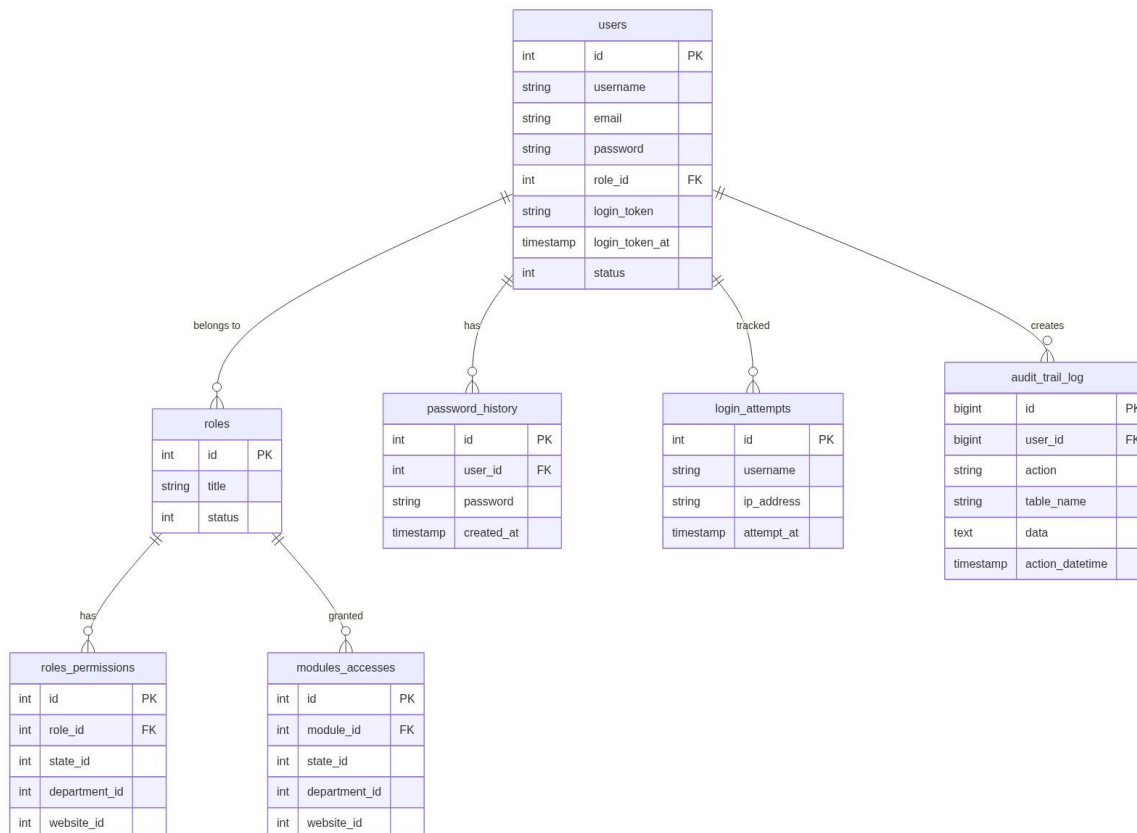


Figure: 05_database_er_diagram Diagram 2

3. Audit Reports Domain

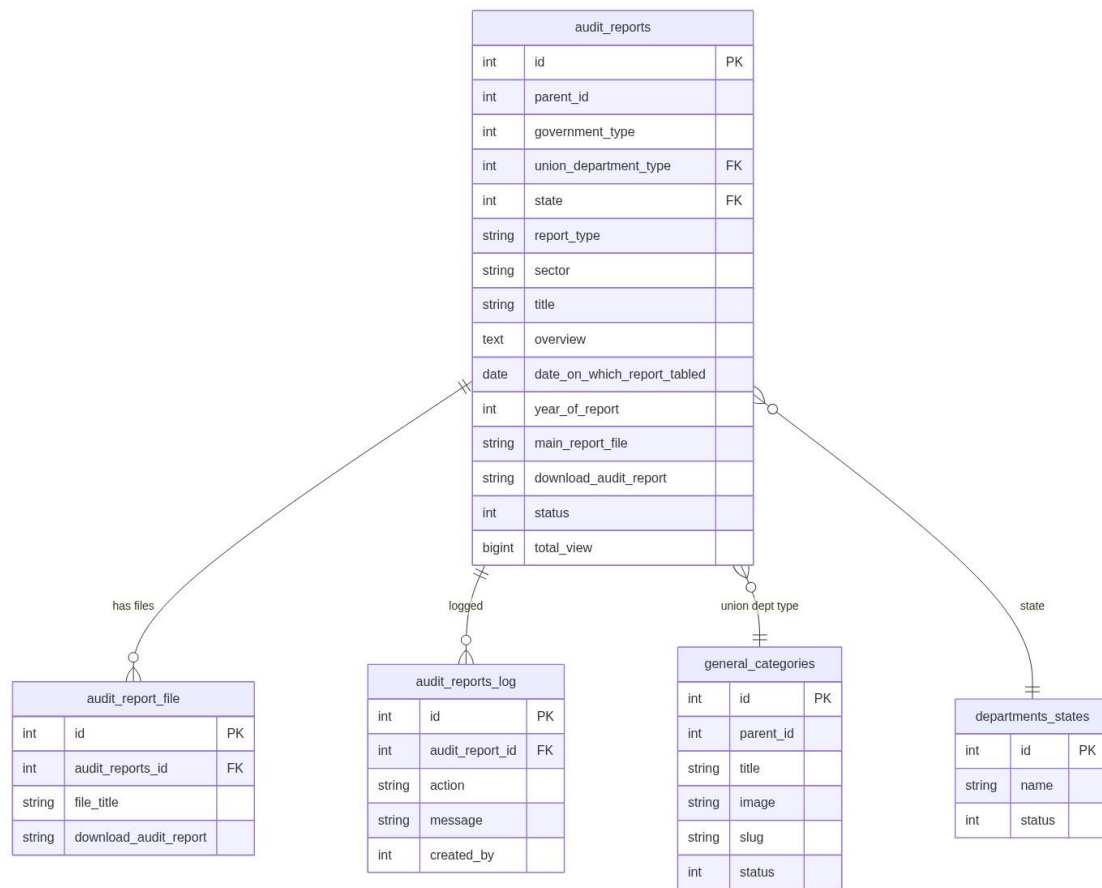


Figure: 05_database_er_diagram Diagram 3

4. Circulars & Notices Domain

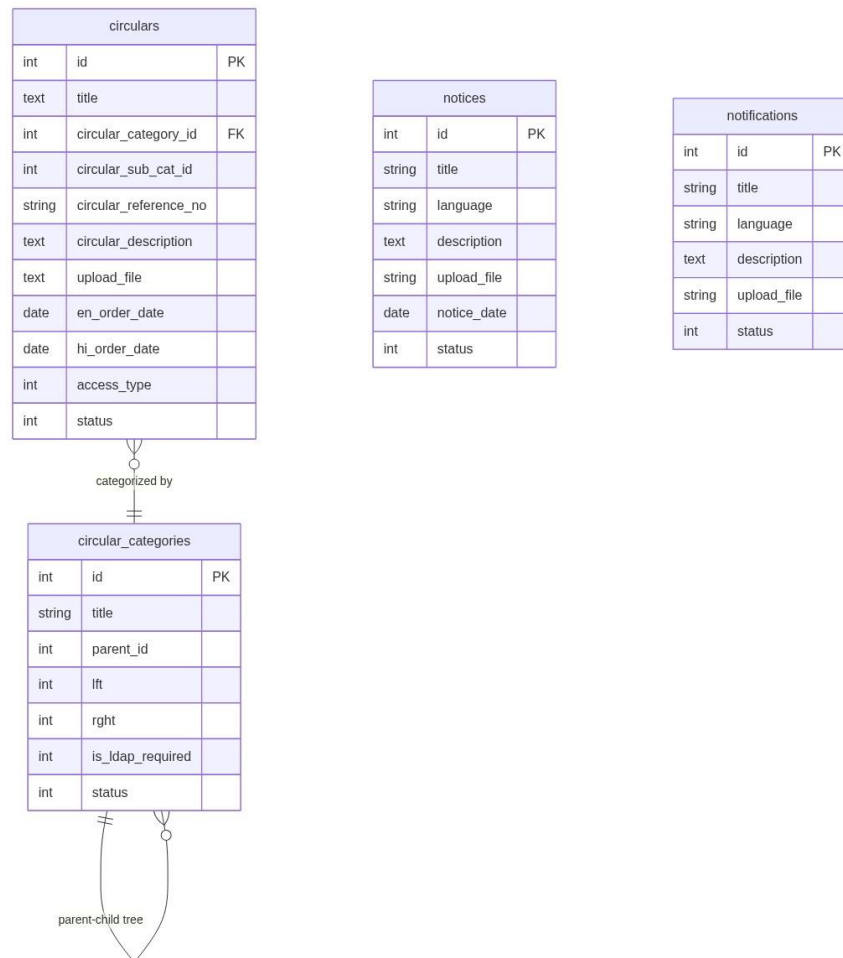


Figure: 05_database_er_diagram Diagram 4

5. ICISA Training Institute Domain

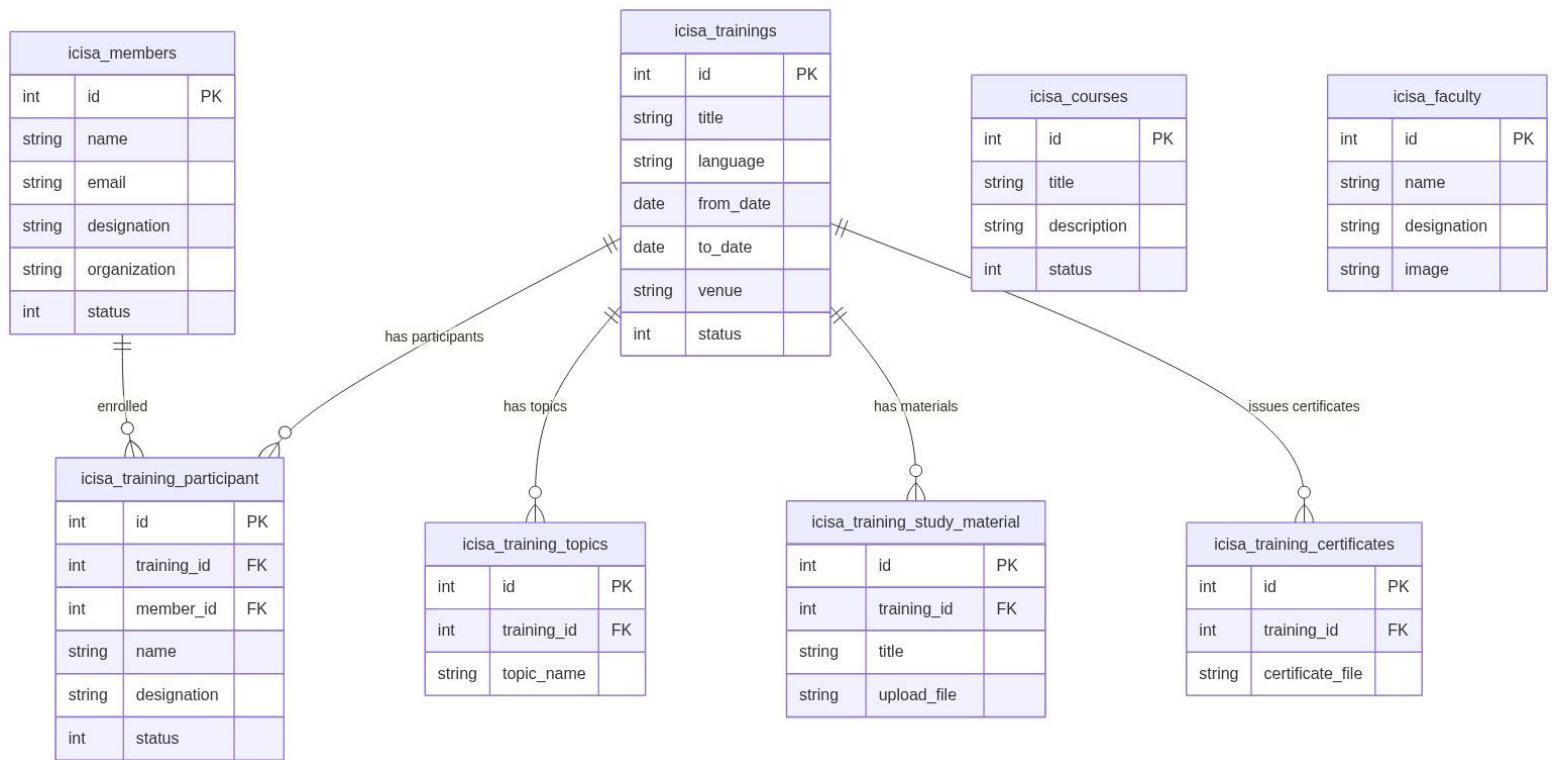


Figure: 05_database_er_diagram Diagram 5

6. Examination Domain

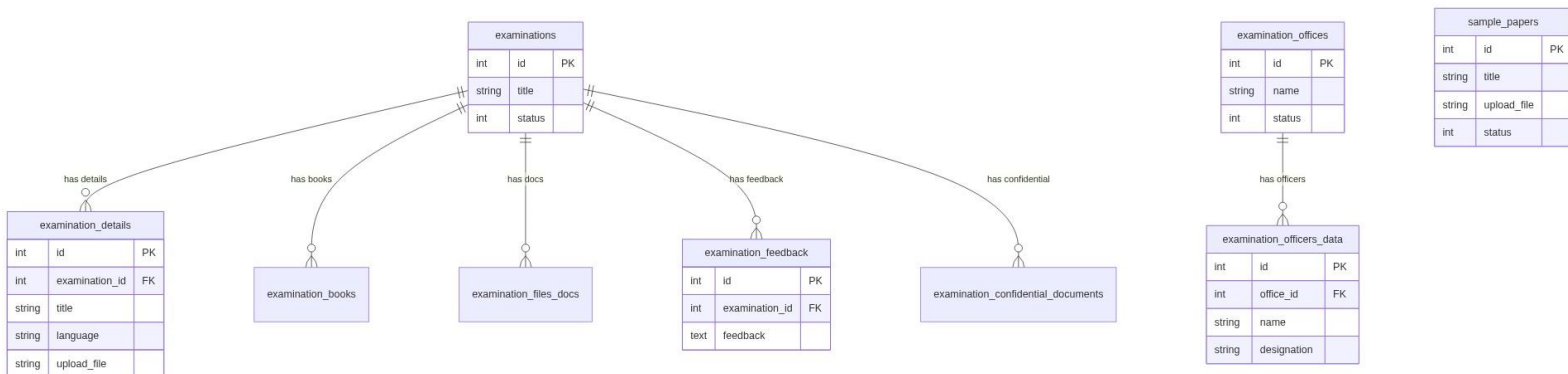


Figure: 05_database_er_diagram Diagram 6

7. Menu & Navigation Domain

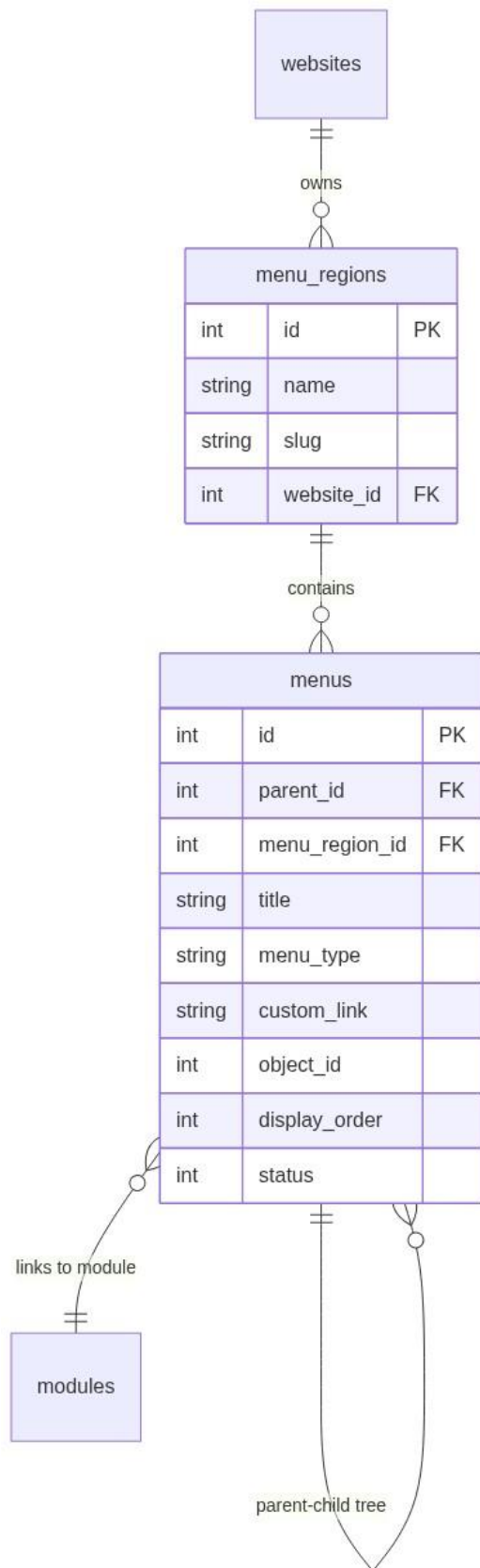


Figure: 05_database_er_diagram Diagram 7

8. Media & Gallery Domain

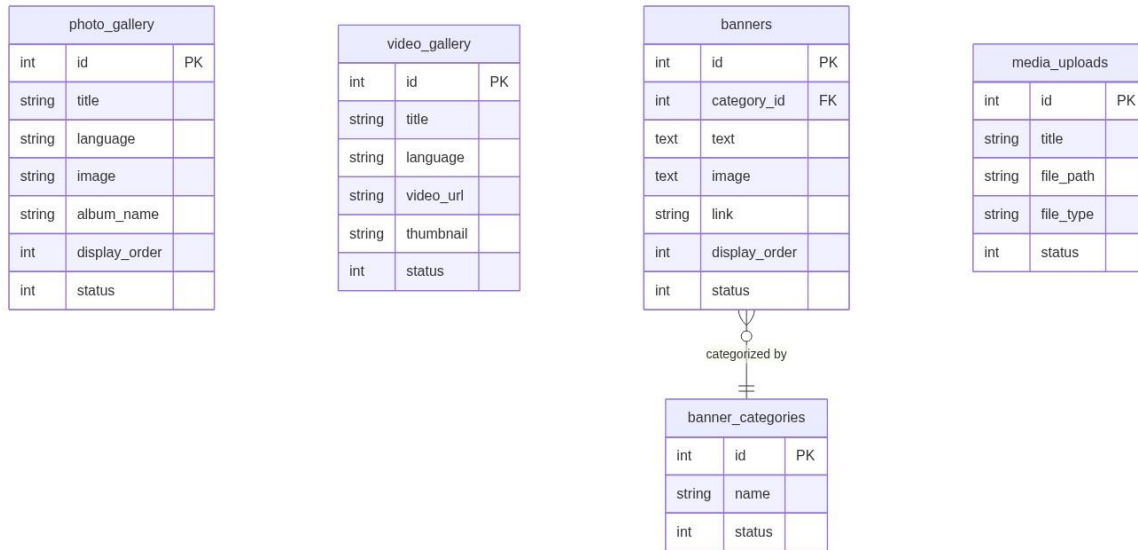


Figure: 05_database_er_diagram Diagram 8

9. HR & Staff Management Domain

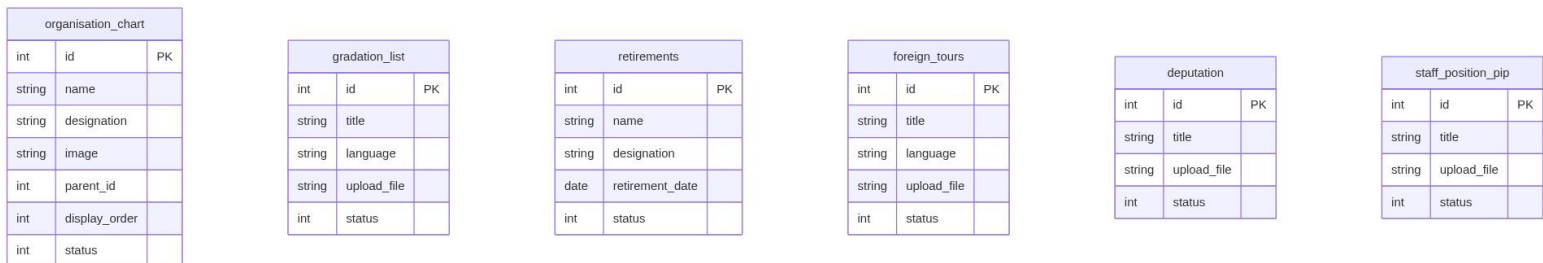


Figure: 05_database_er_diagram Diagram 9

10. Public Interaction Domain

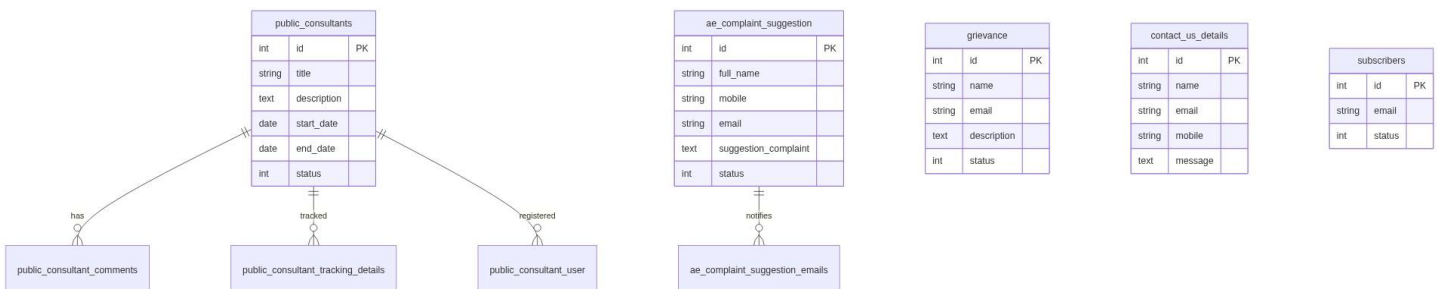


Figure: 05_database_er_diagram Diagram 10

11. RTI & Publications Domain

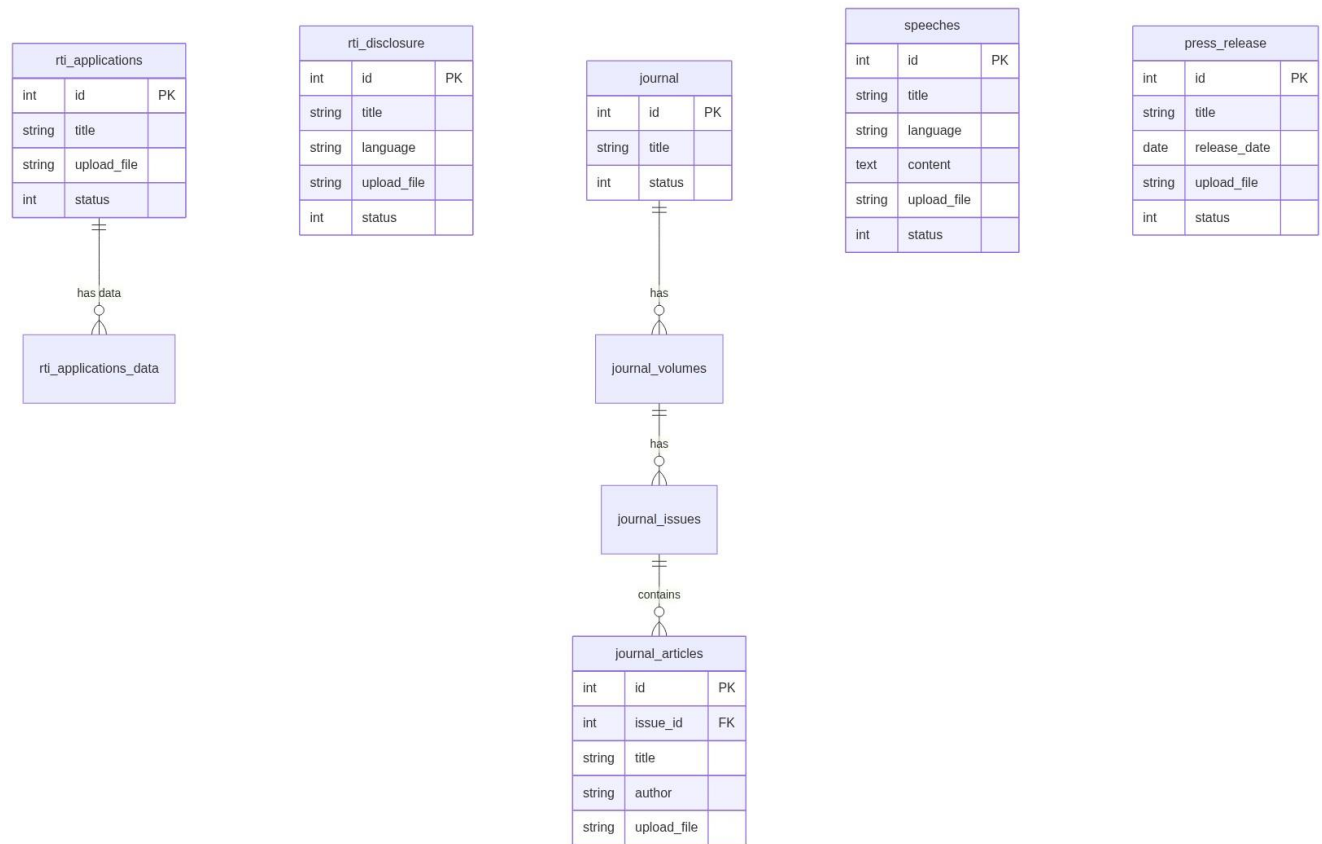


Figure: 05_database_er_diagram Diagram 11

12. Complete Table List (All 199 Tables)

S.no.	Table Name	Domain
1	act_and_mannual	Content
2	administrative_information	Content
3	ae_circulars_office_orders	AE Content
4	ae_complaint_suggestion	Public Interaction
5	ae_complaint_suggestion_emails	Public Interaction
6	ae_notices	AE Content
7	ae_state_accounts	AE Content
8	ae_training	AE Training
9	ag_other_reports	Reports
10	alumni	ICISA
11	applicant_performa_desgination	HR
12	applicant_proforma	HR
13	audit_report_file	Audit
14	audit_reports	Audit
15	audit_reports_log	Audit
16	audit_reports_new	Audit
17	audit_trail_log	System
18	banner_categories	Media
19	banners	Media
20	banners_old	Media (Legacy)
21	bihar_pensions	Pension
22	bihar_pensions_query	Pension
23	blue_economy_center	Content
24	board_committees	Content

25	booking_hostel_rooms	ICISA
26	budget	Content
27	categories	System
28	circular_categories	Circulars
29	circulars	Circulars
30	collaborations	Content
31	combined_accounts	Reports
32	comments	Public Interaction
33	contact_us_details	Public Interaction
34	contact_us_oios_details	Public Interaction
35	content_accesses	System
36	core_published_papers	Publications
37	countries	System
38	course_contents	ICISA
39	course_material	ICISA
40	current_programme	ICISA
41	department_gradation_list	HR
42	departments	System
43	departments_states	System
44	deputation	HR
45	designation_hierarchy	HR
46	districts	System
47	drawing_disb_officers	HR
48	events	Content
49	exam_faqs	Examination
50	exam_wing_circulars	Examination
51	examination_books	Examination
52	examination_confidential_documents	Examination
53	examination_data_collection_forms	Examination
54	examination_details	Examination
55	examination_feedback	Examination
56	examination_files_docs	Examination
57	examination_officers_data	Examination
58	examination_offices	Examination
59	examinations	Examination
60	examinations_wings	Examination
61	existing_policies_transfer_posting_group_bc_staff	HR Policy
62	faqs	Content
63	fields	System (Forms)
64	footer_slider_items	Media
65	foreign_tours	HR
66	form_field_relations	System (Forms)
67	former_cag	Content
68	forms	System (Dynamic Forms)
69	ge	Content
70	general_categories	System
71	gpf	Content
72	gradation_list	HR
73	grievance	Public Interaction
74	guidance_notes_practice_guides	Content
75	guidelines	Content
76	home_blocks	CMS
77	home_regions	CMS
78	home_regions_blocks	CMS
79	hostel_feedback	ICISA

80	iaad_kms_content	KMS
81	iaas_officers_trainee	HR
82	icisa_activities	ICISA
83	icisa_circulars	ICISA
84	icisa_courses	ICISA
85	icisa_deputation_and_notices	ICISA
86	icisa_email_validation_list	ICISA
87	icisa_faculty	ICISA
88	icisa_it_audit_reports	ICISA
89	icisa_marquee	ICISA
90	icisa_members	ICISA
91	icisa_members_feedback	ICISA
92	icisa_resources	ICISA
93	icisa_stm_research_project	ICISA
94	icisa_study_material	ICISA
95	icisa_training_certificates	ICISA
96	icisa_training_participant	ICISA
97	icisa_training_study_material	ICISA
98	icisa_training_topics	ICISA
99	icisa_trainings	ICISA
100	icisa_virtual_publishing	ICISA
101	immovable_property_returns	HR
102	important_dates	Content
103	important_links	Content
104	in_house_systems	Content
105	journal	Publications
106	journal_articles	Publications
107	journal_issues	Publications
108	journal_mgmt_training	Publications
109	journal_volumes	Publications
110	kms_categories	KMS
111	languages	System
112	latest_updates	Content
113	library	ICISA
114	login_attempts	System
115	manuals	Content
116	media_uploads	Media
117	menu_groups	Navigation
118	menu_regions	Navigation
119	menus	Navigation
120	modules	System
121	modules_accesses	System
122	news	Content
123	news_letter	Content
124	note_display_text	System
125	notices	Notices
126	notifications	Notices
127	office_mannual	Content
128	office_name_list	HR
129	old_audit_reports	Audit (Legacy)
130	old_audit_reports_log	Audit (Legacy)
131	org_charge_master	Organisation
132	org_reporting_to	Organisation
133	organisation_chart	Organisation
134	outstanding_treasury_inspection_report	Reports

135	page_translations	CMS
136	pages	CMS
137	password_history	Auth
138	pension	Content
139	performance_activity_report	Reports
140	photo_gallery	Media
141	policy_and_guidelines	Content
142	posting_transfer_guidelines	HR Policy
143	press_clipping	Publications
144	press_release	Publications
145	public_consultant_comments	Public Interaction
146	public_consultant_tracking_details	Public Interaction
147	public_consultant_user	Public Interaction
148	public_consultants	Public Interaction
149	qr_codes	System
150	quick_links	Content
151	rajbhasha_cadre	HR
152	recruitment_notices	HR
153	recruitment_rules	HR
154	research_paper	Publications
155	resource_persons	ICISA
156	retirements	HR
157	roles	Auth
158	roles_permissions	Auth
159	rti_applications	RTI
160	rti_applications_data	RTI
161	rti_disclosure	RTI
162	rti_journal	RTI
163	sample_papers	Examination
164	silver_form_model	System
165	speeches	Publications
166	staff_associations_jcm	HR
167	staff_position_pip	HR
168	state_accounts_report	Reports
169	state_data	System
170	states	System
171	status_of_audit_reports	Reports
172	student_internship_programme	HR
173	study_reports	Reports
174	subscribers	Public Interaction
175	subsite_org_charge_master	Organisation
176	subsite_org_desig_hierarchy	Organisation
177	subsite_whats_new	Content
178	subsites_org_struct	Organisation
179	template_categories	System
180	templates	System
181	tenders	Tenders
182	tour_programme	HR
183	training	ICISA
184	training_calendar	ICISA
185	training_events	ICISA
186	treasury_officers	HR
187	trigger_organisation_chart	Organisation
188	user_offices	Auth
189	users	Auth

190	video_gallery	Media
191	visitors	System
192	websites	System
193	welfare	HR
194	whats_new	Content
195	wings	System
196	young_professional_programme	HR
197	bihar_pensions	Pension
198	bihar_pensions_query	Pension
199	hostel_feedback	ICISA

06 — Routing & Page Resolution

1. Route Architecture Overview

The application has **5 route prefixes** defined in [routes.php](#):

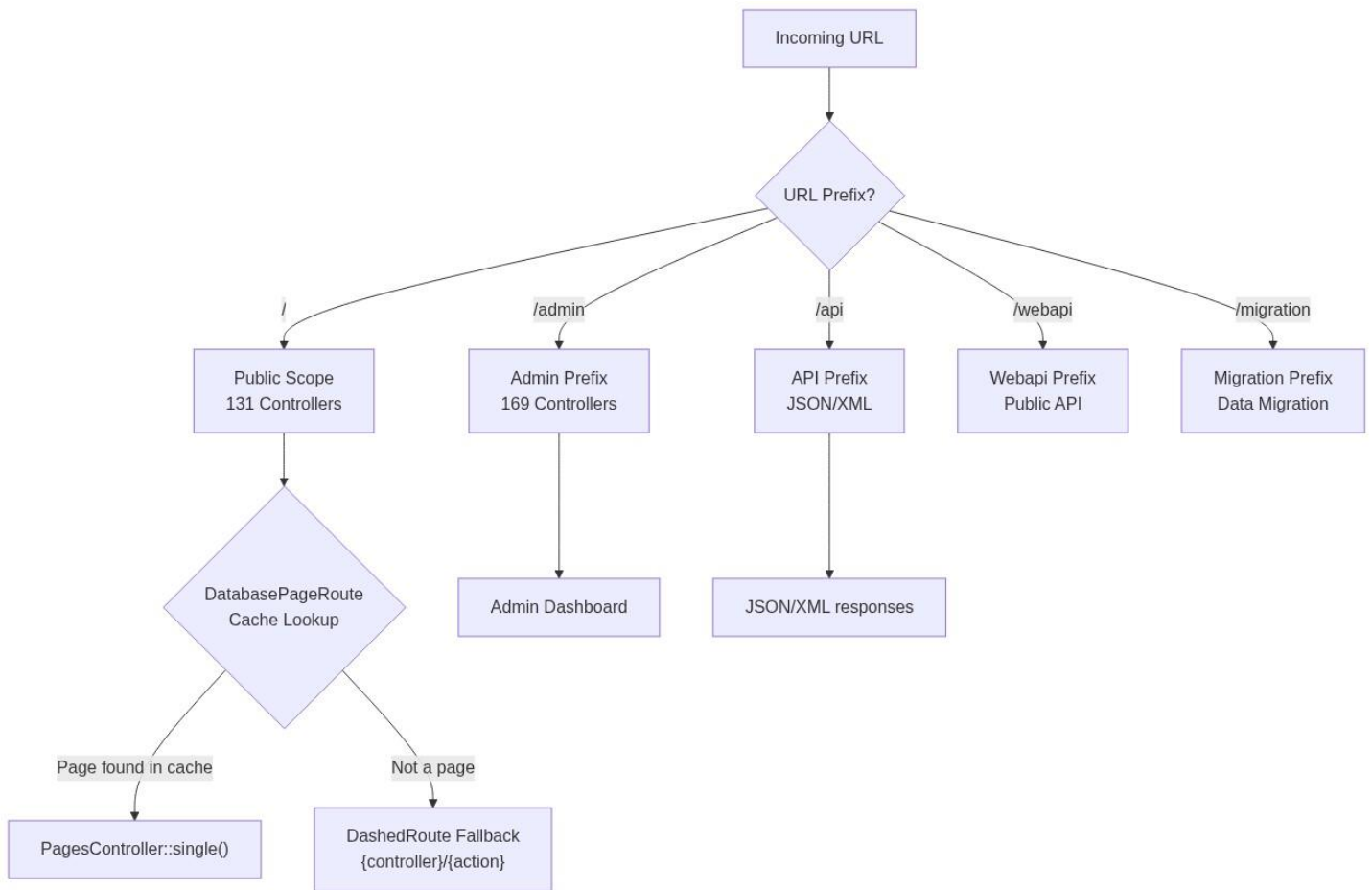


Figure: 06_routing_and_pages Diagram 1

2. URL Structure Patterns

The URL scheme supports combinations of **website slugs** and **language codes**:

Pattern	Example	Resolution
/	cag.gov.in/	HomeController::index()
/ {language}	/en, /hi	HomeController::index() + locale
/ {website}	/naaa	HomeController::index() for sub-site
/ {website}/ {language}	/naaa/en	Sub-site homepage with locale
/ {website}/ {language}/ {controller}/ {action}	/naaa/en/circulars/index	Standard controller route
/ {language}/ {page_slug}	/en/about-us	PagesController::single() ()
/ {website}/ * (catch-all)	/naaa/any-slug	DatabasePageRoute lookup

Special Multi-Segment Website Slugs

Some sub-sites have **multi-segment URLs** (e.g., `rly/mumbai-i`, `cen/new-delhi-iv`). These are hard-coded in the `getMismatchedUrl()` function:

```
/cen/new-delhi-iv, /defence/new-delhi-i, /defence/new-delhi-ii,  
/rly/new-delhi-i, /rly/new-delhi-ii, /rly/kolkata-iii,  
/rly/kolkata-ii, /cen/new-delhi-ii, /mab/kolkata-ii,  
/mab/new-delhi-iii, /mab/new-delhi-iv, /mab/mumbai-ii,  
/cen/new-delhi-iii, /naaa/naaa75years
```

3. DatabasePageRoute — O(1) CMS Page Resolution

File: [DatabasePageRoute.php](#)

This is the most important routing component. It provides **constant-time page lookups** using a pre-built cache.

How It Works

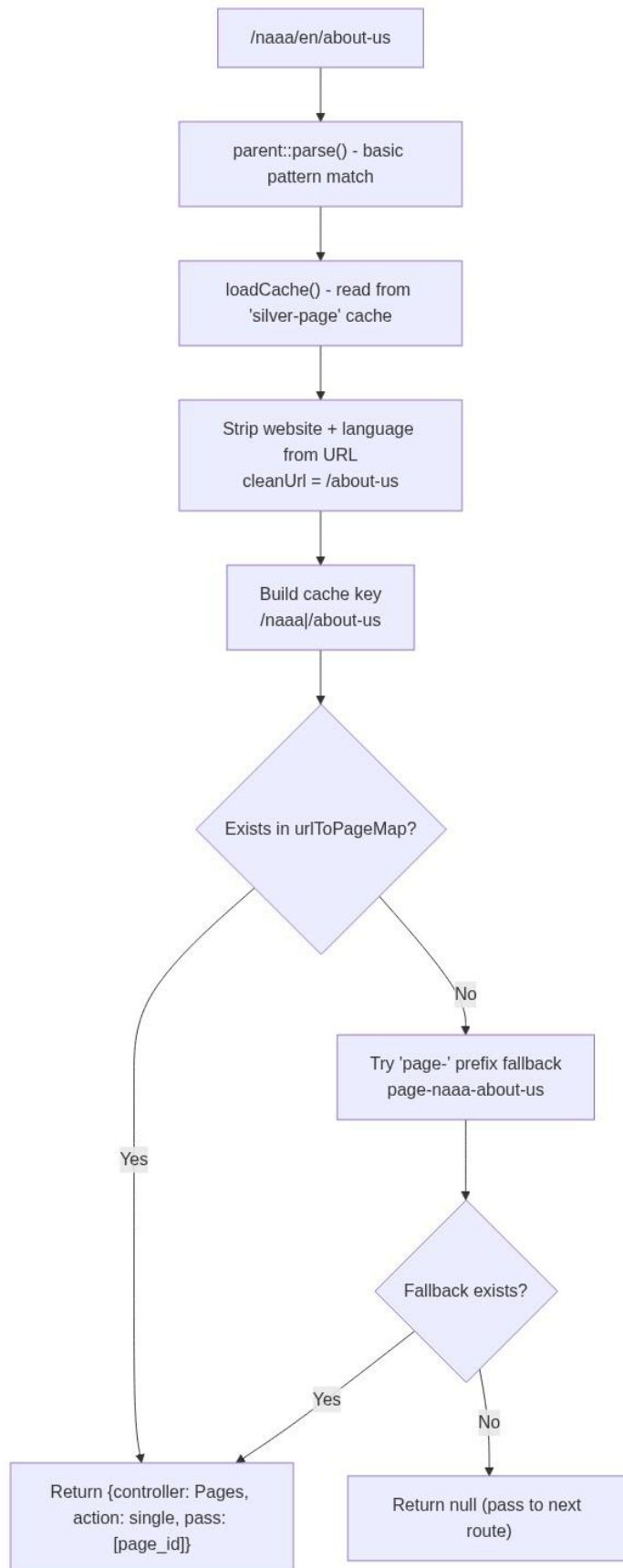


Figure: 06_routing_and_pages Diagram 2

Cache Structure

- **Cache key:** `silverpage` in `silver-page` cache engine
- **Data format:** `$pages[$websiteSlug][$index] = ['id' => ..., 'url' => ..., 'website_id' => ...]`
- **Lookup maps built in `loadCache()`:**
- `$urlToPageMap["websiteSlug|pageUrl"] => page data` — For URL → page resolution
- `$idToPageMap[pageId] => page data` — For page ID → URL generation (reverse routing)

Reverse Routing (match)

The `match()` method handles URL generation from controller params. When a template calls `Router::url(['controller' => 'Pages', 'action' => 'single', 'pageId'], it:`

1. Looks up the page in `$idToPageMap`
2. Prepends the language and website slug
3. Returns the clean URL path

4. PagesController — CMS Page Rendering

File: [PagesController.php](#)

Actions

Action	Purpose
<code>display()</code>	Renders static templates from <code>templates/Pages/</code> (CakePHP default)
<code>home()</code>	Renders the home page for a sub-site. Looks up pages where <code>is_home = 1</code>
<code>single(\$page_id)</code>	The main CMS page renderer. Accepts numeric ID or string slug

`single()` Logic Flow

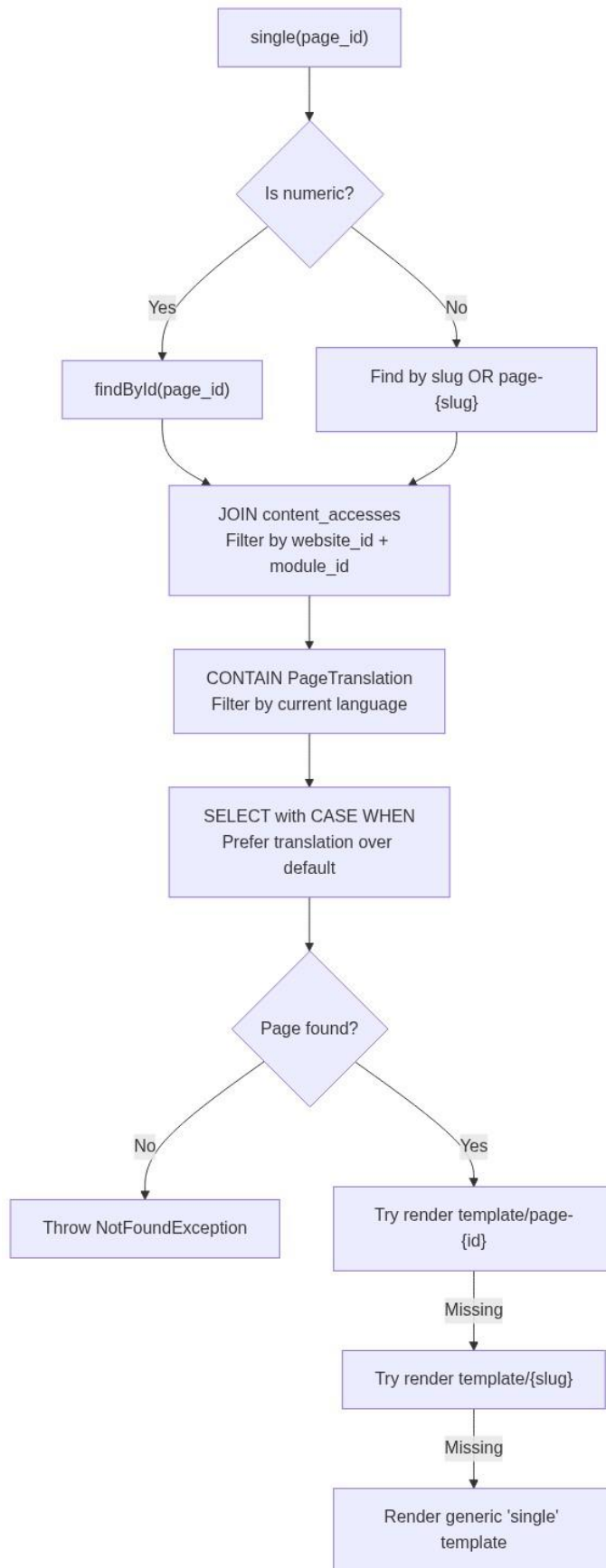


Figure: 06_routing_and_pages Diagram 3

Translation Logic

Pages support multilingual content via `page_translations` table. The SQL uses `CASE WHEN` expressions:

```
CASE WHEN PageTranslation.title != ''  
      THEN PageTranslation.title  
      ELSE Pages.title  
END
```

5. Language Routing

LanguageRoutingMiddleware Logic

File: [LanguageRoutingMiddleware.php](#)

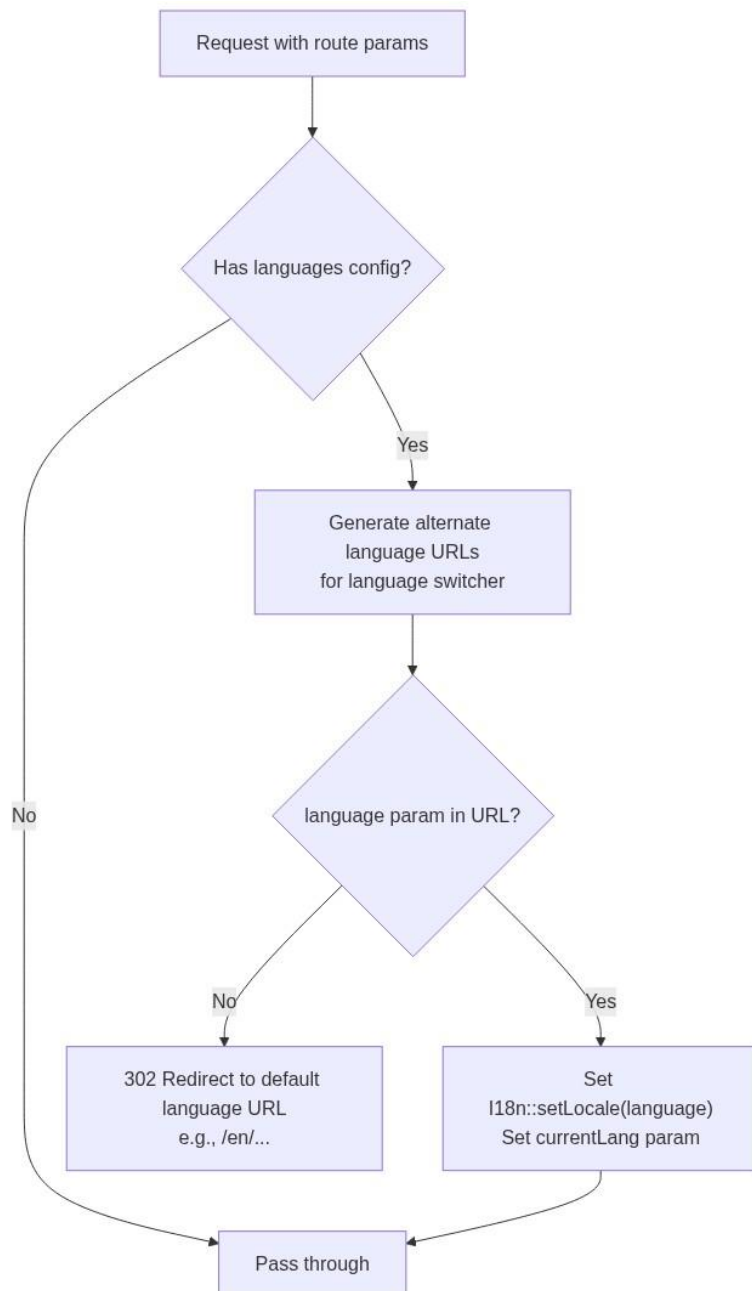


Figure: 06_routing_and_pages Diagram 4

Languages Cache

- Languages are cached in the `language` cache engine
- Loaded from the `languages` table on first request
- Each language has: `id`, `name`, `culture` (e.g., `en`, `hi`), `is_default`

6. Admin Route Prefix

The `/admin` prefix scope mirrors the public scope but:

- Routes to `Admin/` namespaced controllers
- Dashboard is `Admin/DashboardController::index()`
- Adds a **URL filter** that propagates `module_id` and `website_id` query parameters across admin navigation
- Supports all the same website/language combinations

7. Route Priority Order

Routes are evaluated in this order (first match wins):

1. **DatabasePageRoute** with `/ {website} / {language} / *` — Cached CMS pages (O(1))
2. **DatabasePageRoute** with `/ {website} / *` — Cached CMS pages without language
3. **DatabasePageRoute** with `/ {language} / *` — Main site pages with language
4. **DatabasePageRoute** with `/ *` — Main site pages
5. **Fallback page-slug routes** (`page-[^\s/]+` pattern)
6. **Named page routes** (`/pages/single/ *`)
7. **Login route** (`/ {language} / login`)
8. **Multi-segment website routes** (hardcoded `$multi` loop)
9. **Dashboard routes** (`/ {website} / {website} /`, etc.)
10. **Standard DashedRoute** (`/ {website} / {language} / {controller} / {action}`)
11. **Language-only DashedRoute** (`/ {language} / {controller} / {action}`)
12. **CakePHP fallbacks** (`/ {controller} / {action}`)

07 — Public Controllers (131 Files)

All public controllers live in `src/Controller/` and extend `AppController`. They handle the public-facing website pages. By default, all actions are set to `allowUnauthenticated` in `beforeFilter()`.

1. Controller Hierarchy

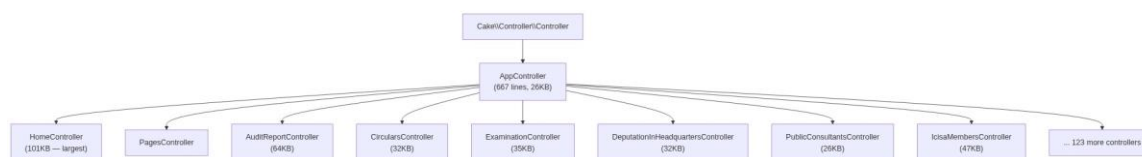


Figure: 07_public_controllers Diagram 1

2. AppController — Base Class (All Public Controllers)

File: `AppController.php` — 667 lines

Components Loaded

- **Flash** — Session flash messages
- **General** — Provides access to `GeneralHelper` and `SilverFormHelper`
- **Language** — Language configuration
- **Email** — Email sending abstraction
- **Menu** — Menu rendering and access control

- `Authentication.Authentication` — CakePHP 5 authentication
- `Auth` — Compatibility shim (bridges CakePHP 3 Auth API)

Key Methods

Method	Purpose
<code>initialize()</code>	Loads components, sets language/website, builds active menu, checks access
<code>beforeFilter()</code>	Marks all public actions as unauthenticated, injects CSRF token, tracks visitors
<code>beforeRender()</code>	Sets theme via <code>viewBuilder()->setTheme()</code> , falls back to <code>Fallback/index</code> template
<code>getModuleId()</code>	Looks up the module ID for the current controller from the <code>modules</code> table
<code>getFrontModuleId()</code>	Gets the front-end module ID for menu highlighting
<code>setLanguage()</code>	Sets the active language from the route param or default
<code>setWebsite()</code>	Determines the current sub-site from the URL and loads its config
<code>visitorStats()</code>	Counts weekly visitors (COEFA theme only)
<code>trackVisitor()</code>	Inserts visitor tracking record

3. Controller Inventory — Grouped by Domain

3.1 Homepage & CMS Pages

Controller	File Size	Key Actions	ORM Tables Used
HomeController	101KB	<code>index</code> , <code>ourOffice</code>	Banners, AuditReports, FooterSliderItems, WhatsNew, News, Events, Tenders, Circulars, HomeBlocks, HomeRegions
PagesController	6KB	<code>display</code> , <code>home</code> , <code>single</code>	Pages, PageTranslation, ContentAccesses
SearchController	4KB	<code>index</code>	Pages, AuditReports

			(UNION search with ILIKE scoring)
SitemapController	747B	index	Pages
HistoryController	575B	index	—
WebCronController	666B	index	—

3.2 Audit Reports & State Accounts

Controller	File Size	Key Actions	ORM Tables Used
AuditReportController	64KB	index, view, download, getStates, filterReports	AuditReports, AuditReportFile, GeneralCategories, DepartmentsStates
OldAuditReportsController	7KB	index, view	OldAuditReports
StateAccountsReportController	10KB	index, view	StateAccountsReport, GeneralCategories
StatusOfAuditReportsController	2KB	index	StatusOfAuditReports
AgOtherReportsController	3KB	index	AgOtherReports, GeneralCategories
PerformanceActivityReportController	3KB	index	PerformanceActivityReport
StudyReportsController	4KB	index	StudyReports
OutstandingTreasuryInspectionReportController	3KB	index	OutstandingTreasuryInspectionReport
CombinedAccountsController	3KB	index	CombinedAccounts

3.3 Circulars, Notices & Notifications

Controller	File Size	Key Actions	ORM Tables Used
CircularsController	32KB	index, view, newCircular, getCircularSubCategories, checkAccess	Circulars, CircularCategories, ContentAccesses
NoticeController	3KB	index	Notices
NotificationController	2KB	index	Notifications
WhatsNewController	4KB	index	WhatsNew
NewsController	1KB	index, view	News
NewsLetterController	2KB	index	NewsLetter
AeNoticesController	2KB	index	AeNotices
AeCircularsOfficeOrdersController	5KB	index	AeCircularsOfficeOrders, GeneralCategories
ExamWingCircularsController	3KB	index	ExamWingCirculars

3.4 Examination Domain

Controller	File Size	Key Actions	ORM Tables Used
ExaminationController	35KB	index, view, singleExam, scheduleList, download	Examinations, ExaminationDetails, ExaminationBooks, ExaminationFeedback
ExamFaqsController	1KB	index	ExamFaqs
SamplePapersController	3KB	index	SamplePapers

3.5 ICISA Training Institute (20 Controllers)

Controller	File Size	Key Actions	ORM Tables Used
IcisaTrainingsController	12KB	index, view, pastTraining	IcisaTrainings, IcisaTrainingParticipant
IcisaMembersController	47KB	index, register, login, profile, editProfile, changePassword	IcisaMembers, IcisaEmailValidationList
IcisaFacultyController	2KB	index	IcisaFaculty
IcisaStudyMaterialController	3KB	index	IcisaStudyMaterial
IcisaCircularsController	2KB	index	IcisaCirculars
IcisaCoursesController	2KB	index	IcisaCourses
IcisaActivitiesController	2KB	index	IcisaActivities
IcisaDeputationAndNoticesController	2KB	index	IcisaDeputationAndNotices
IcisaResourcesController	6KB	index	IcisaResources
IcisaItAuditReportsController	5KB	index	IcisaItAuditReports
IcisaVirtualPublishingController	3KB	index	IcisaVirtualPublishing
IcisaStmResearchProjectController	1KB	index	IcisaStmResearchProject
IcisaTrainingParticipantController	5KB	index	IcisaTrainingParticipant
IcisaTrainingCertificatesController	2KB	index	IcisaTrainingCertificates
IcisaMembersFeedbackController	1KB	add	IcisaMembersFeedback
IcisaHomeSearchController	3KB	index	Multiple ICISA tables (search)

3.6 HR & Staff Management

Controller	File Size	Key Actions	ORM Tables Used
OrganisationChartController	9KB	index, view	OrganisationChart
GradationListController	5KB	index	GradationList
RetirementsController	2KB	index	Retirements
DeputationController	2KB	index	Deputation

DeputationInHeadquartersController	32KB	index, view, apply, download	Deputation, Users
ForeignToursController	3KB	index	ForeignTours
TourProgrammeController	3KB	index	TourProgramme
StaffPositionPipController	3KB	index	StaffPositionPip
StaffAssociationsJcmController	3KB	index	StaffAssociationsJcm
DepartmentGradationListController	4KB	index	DepartmentGradationList
DrawingDisbOfficersController	2KB	index	DrawingDisbOfficers
TreasuryOfficersController	2KB	index	TreasuryOfficers
ImmovablePropertyReturnsController	2KB	index	ImmovablePropertyReturns
WelfareController	2KB	index	Welfare
ApplicantProformaController	7KB	add, view, download	ApplicantProforma
ApplicantProformaDesignationController	2KB	index	ApplicantPerformaDesgination

3.7 Recruitment & Career

Controller	File Size	Key Actions	ORM Tables Used
RecruitmentNoticesController	5KB	index	RecruitmentNotices
RecruitmentRulesController	4KB	index	RecruitmentRules
StudentInternshipProgrammeController	3KB	index, apply	StudentInternshipProgramme
YoungProfessionalProgrammeController	3KB	index, apply	YoungProfessionalProgramme

3.8 Publications & Media

Controller	File Size	Key Actions	ORM Tables Used
SpeechesController	3KB	index, view	Speeches
PressReleaseController	4KB	index, view	PressRelease
PressClippingController	3KB	index	PressClipping
JournalController	3KB	index, view	Journal, JournalVolumes, JournalIssues
JournalArticlesController	4KB	index, view	JournalArticles
JournalMgmtTrainingController	2KB	index	JournalMgmtTraining
ResearchPaperController	2KB	index	ResearchPaper
PhotoGalleryController	6KB	index, view, albumPhotos	PhotoGallery
VideoGalleryController	4KB	index	VideoGallery

3.9 Public Interaction & RTI

Controller	File Size	Key Actions	ORM Tables Used
PublicConsultantsController	26KB	index, view, addComment, register, trackStatus	PublicConsultants, PublicConsultantComments, PublicConsultantUser, PublicConsultantTrackingDetails
GrievanceController	4KB	index, add	Grievance
ContactUsDetailsController	16KB	index, add	ContactUsDetails
ContactUsOiosDetailsController	25KB	index, add	ContactUsOiosDetails
CommentsController	8KB	add	Comments
SubscribersController	5KB	subscribe, unsubscribe	Subscribers
AeComplaintSuggestionController	8KB	index, add, view, track	AeComplaintSuggestion, AeComplaintSuggestionEmails
RtiApplicationsController	2KB	index	RtiApplications
RtiDisclosureController	4KB	index	RtiDisclosure
RtiJournalController	2KB	index	RtiJournal

3.10 Content & Policy Pages

Controller	File Size	Key Actions	ORM Tables Used
ActAndMannualController	2KB	index	ActAndMannual
AdministrativeInformationController	2KB	index	AdministrativeInformation
BudgetController	2KB	index	Budget
GuidelinesController	2KB	index	Guidelines
GuidanceNotesPracticeGuidesController	2KB	index	GuidanceNotesPracticeGuides
ManualsController	2KB	index	Manuals
OfficeMannualController	2KB	index	OfficeMannual
PolicyAndGuidelinesController	2KB	index	PolicyAndGuidelines
FormerCagController	1KB	index	FormerCag
FagsController	2KB	index	Fags
ImportantLinksController	2KB	index	ImportantLinks
EventsController	5KB	index, view	Events
TendersController	4KB	index, view	Tenders
BoardCommitteesController	2KB	index	BoardCommittees
CollaborationsController	2KB	index	Collaborations
InHouseSystemsController	1KB	index	InHouseSystems
BlueEconomyCenterController	3KB	index	BlueEconomyCenter

3.11 AE (Accountant General) Subsites

Controller	File Size	Key Actions
AeStateAccountsController	4KB	index
AeTrainingController	1KB	index

GeController	3KB	index
GpfController	2KB	index
PensionController	2KB	index
BiharPensionsController	3KB	index, search
BiharPensionsQueryController	3KB	index, search

3.12 Training & Calendar

Controller	File Size	Key Actions
TrainingController	1KB	index
TrainingCalendarController	3KB	index
TrainingEventsController	2KB	index
CurrentProgrammeController	2KB	index
CourseContentsController	1KB	index
CourseMaterialController	2KB	index
ResourcePersonsController	1KB	index
LibraryController	2KB	index

3.13 Organisation & Subsite Structure

Controller	File Size	Key Actions
SubsitesOrgStructController	15KB	index, view, orgChart
SubsiteWhatsNewController	2KB	index

3.14 Authentication & SSO

Controller	File Size	Key Actions
ParichaySsoController	10KB	login, callback, logout
AlumniController	3KB	index, register
AlumniDirectoryController	5KB	index

3.15 KMS (Knowledge Management) & IAAD

Controller	File Size	Key Actions
IaadKmsContentController	5KB	index, view

3.16 Utility & Supporting Controllers

Controller	File Size	Key Actions
AjaxController	24KB	getMostViewRecent , getStates , various AJAX endpoints
ErrorController	2KB	CakePHP error handling
ViewFileController	3KB	index — secure file serving
FilesController	926B	File serving
UserOfficesController	1KB	index

3.17 Policies & HR Rules

Controller	File Size	Key Actions
ExistingPoliciesTransferPostingGroupBCStaffController	3KB	index
PostingTransferGuidelinesCommercialSaosAaosController	3KB	index
RajbhashaCadreController	—	index (Hindi language cadre)

4. Common Controller Pattern

Most public controllers follow this standard pattern:

```
class SomeController extends AppController
{
    public function initialize(): void
    {
        parent::initialize();
        $this->Auth->allow();           // All actions public
        $this->language = $this->request->getParam('language');
    }

    public function index()
    {
        $moduleId = $this->getModuleId('SomeModule');
        $conditions = ['status' => 1, 'language' => $this->language];

        $items = $this->SomeTable->find('all')
            ->join([                    // Content access control
                'table' => 'content_accesses',
                'type' => 'INNER',
                'conditions' => 'content_accesses.content_id = "SomeTable".id
            ]
        );
    }
}
```

```
        AND content_accesses.module_id = ' . $moduleId . '
        AND content_accesses.website_id = ' . $this->websiteId,
    ])
    ->where($conditions)
    ->order(['display_order' => 'ASC'])
    ->toArray();

    $this->set(compact('items'));
}
}
```


08 — Admin Controllers (169 Files)

All admin controllers live in `src/Controller/Admin/` and extend `Admin/AppController`. They handle the CMS back-office (content management, user administration, configuration).

1. Admin AppController — Base Class (935 Lines, 41KB)

File: [Admin/AppController.php](#)

Key Differences from Public AppController

Feature	Public AppController	Admin AppController
Auth session key	Auth	Auth.Admin
Login redirect	/ (Home)	/admin/dashboard
Default layout	default (public)	admin
IP/UA binding	No	Yes — dies on mismatch
Role checking	Menu-level only	Full RBAC per module
Token validation	No	Yes — <code>isTokenValid()</code>
Referer checking	No	Yes — blocks cross-origin POST

Security Features in Admin AppController

- Referer Validation:** On POST/PUT/PATCH requests, checks that `$request->referer()` hostname matches the app's base URL
- IP Address Binding:** Compares `$_SERVER['REMOTE_ADDR']` to `Session('Auth.ipaddress')`; destroys session on mismatch
- User-Agent Binding:** Compares `$_SERVER['HTTP_USER_AGENT']` to `Session('Auth.useragent')`; destroys session on change
- Cookie Validation:** Decrypts and validates a secure cookie against the session identity
- Token Validation:** `isTokenValid()` checks for valid session tokens to prevent replay attacks
- ICISA Member Redirect:** Users with `ICISA_MEMBER_ROLE_ID` are redirected to `/icisa`
- Action/Data ID Tracking:** For `view` and `delete` actions, stores the data ID in session to prevent parameter tampering

Role-Based Module Access

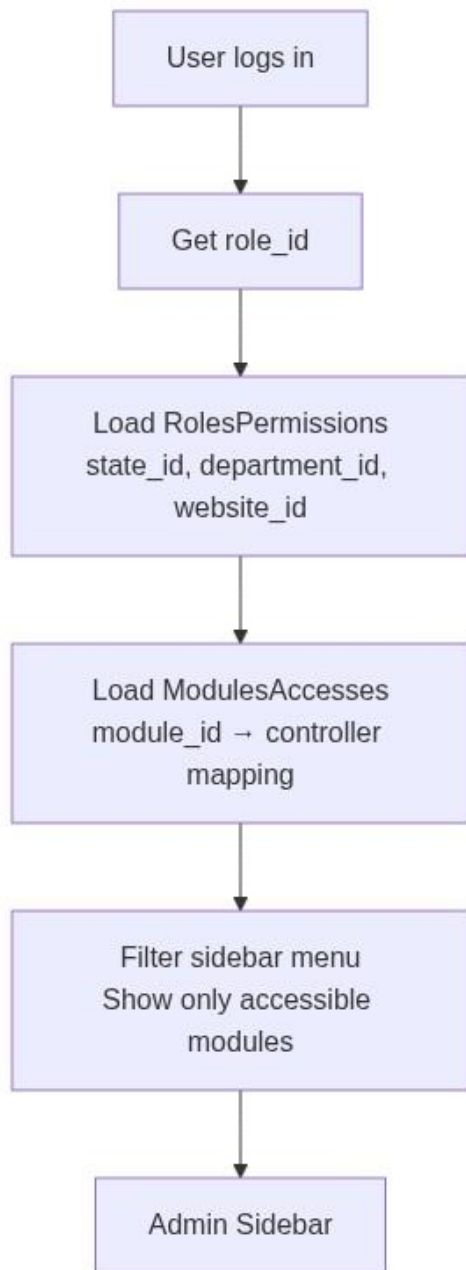


Figure: 08_admin_controllers Diagram 1

2. Standard Admin CRUD Pattern

Every admin controller follows this pattern with 5–7 standard actions:

```
class SomeController extends ApplicationController
{
    public function initialize(): void
```

```

{
    parent::initialize();
    $this->web['controller'] = $this->request->getParam('controller');
    $this->web['model'] = 'SomeTable';
}

// LIST with pagination, search, filters
public function index() { ... $this->paginate($query) ... }

// VIEW single record
public function view($id = null) { ... ->get($id) ... }

// CREATE new record (form + POST)
public function add() { ... ->newEntity() -> patchEntity() -> save() ... }

// UPDATE existing record
public function edit($id = null) { ... ->get($id) -> patchEntity() -> save()
... }

// SOFT DELETE (set status = 0)
public function delete($id = null) { ... ->updateQuery()->set(['status' => 0])
... }

// EXPORT to CSV/Excel/PDF
public function export() { ... PhpSpreadsheet or mPDF ... }
}

```

3. Admin Controller Inventory — Grouped by Domain

3.1 System Administration

Controller	Size	Actions	Purpose
UsersController	104KB	login, logout, index, add, edit, view, delete, forgotPassword, resetPassword, changePassword, myProfile, captcha, jcrypto	Full user management + authentication
RolesController	22KB	index, add, edit, view, delete, permissions	Role management with permission assignment
ModulesController	13KB	index, add, edit, delete	Module registration for RBAC
WebsitesController	24KB	index, add, edit, view, delete	Sub-site management
MenusController	22KB	index, add, edit, delete, reorder	Navigation menu CRUD + drag-drop reorder

MenuRegionsController	12KB	index, add, edit, delete	Menu region management
LanguagesController	7KB	index, add, edit, delete	Language configuration
DashboardController	656B	index	Admin landing page
AuditTrailLogController	5KB	index, view	View system audit logs
WingsController	9KB	index, add, edit, delete	Organisational wings
TemplateCategoriesController	7KB	index, add, edit, delete	Template category management
TemplatesController	8KB	index, add, edit, delete	HTML templates
NoteDisplayTextController	10KB	index, add, edit, delete	System notification text

3.2 CMS Content Management

Controller	Size	Actions	Purpose
PagesController	34KB	index, add, edit, view, delete, translate	CMS page management + translations
HomeBlocksController	22KB	index, add, edit, delete	Homepage content blocks
HomeRegionsController	23KB	index, add, edit, delete	Homepage layout regions
QuickLinksController	14KB	index, add, edit, delete	Quick links section
ImportantLinksController	22KB	index, add, edit, delete	Important links
ImportantDatesController	17KB	index, add, edit, delete	Important dates
FagsController	20KB	index, add, edit, delete	FAQ management
WhatsNewController	12KB	index, add, edit, delete	What's New ticker
SubsiteWhatsNewController	20KB	index, add, edit, delete	Sub-site what's new

3.3 Audit Reports Management

Controller	Size	Actions	Purpose
AuditReportsController	85KB	index, add, edit, view, delete, export, importPdfText, bulkUpload	Full audit report lifecycle
OldAuditReportsController	43KB	index, add, edit, view, delete	Legacy audit reports
StatusOfAuditReportsController	15KB	index, add, edit, delete	Audit report status tracking
StateAccountsReportController	26KB	index, add, edit, delete	State accounts reports

3.4 Circulars & Publications

Controller	Size	Actions	Purpose
CircularsController	96KB	index, add, edit, view, delete, export, import	Circular management (largest admin controller)
CircularCategoriesController	25KB	index, add, edit, delete	Circular category tree
SpeechesController	26KB	index, add, edit, delete	Speeches
PressReleaseController	26KB	index, add, edit, delete	Press releases
PressClippingController	21KB	index, add, edit, delete	Press clippings
NewsController	20KB	index, add, edit, delete	News articles
NewsLetterController	21KB	index, add, edit, delete	Newsletters

3.5 HR & Organisation (20+ Controllers)

Controller	Size	Purpose
OrganisationChartController	136KB	Organisation chart (largest file in project)
GradationListController	21KB	Gradation list
RetirementsController	22KB	Retirements
DeputationController	20KB	Deputation management
ForeignToursController	22KB	Foreign tour management
TourProgrammeController	23KB	Tour programme
StaffPositionPipController	21KB	Staff position/PIP
IaasOfficersTraineeController	23KB	IAAS officers/trainees
DepartmentsStatesController	13KB	Departments & states
SubsitesOrgStructController	28KB	Sub-site org structure
SubsiteOrgChargeMasterController	18KB	Sub-site charge master
SubsiteOrgDesigHierarchyController	21KB	Sub-site designation hierarchy

3.6 ICISA Administration (18 Controllers)

Controller	Size	Purpose
IcisaTrainingsController	29KB	Training program management
IcisaTrainingParticipantController	26KB	Participant enrollment
IcisaStudyMaterialController	31KB	Study material upload
IcisaPhotoGalleryController	35KB	Photo gallery management
IcisaMembersController	17KB	Member management
IcisaFacultyController	16KB	Faculty management
Other ICISA Controllers	11-18KB	Activities, Circulars, Courses,

		Deputation, Resources, etc.
--	--	-----------------------------

3.7 Examination Administration (8 Controllers)

Controller	Size	Purpose
ExaminationDataCollectionFormsController	43KB	Exam data collection forms
ExaminationOfficesController	27KB	Exam office management
ExaminationDetailsController	20KB	Exam detail management
ExaminationBooksController	18KB	Exam books
ExaminationConfidentialDocumentsController	16KB	Confidential exam docs
ExaminationsController	9KB	Examination CRUD
ExaminationsWingsController	18KB	Exam wings
SamplePapersController	24KB	Sample papers

3.8 Media & Gallery

Controller	Size	Purpose
BannersController	22KB	Banner image management
BannerCategoriesController	7KB	Banner categories
PhotoGalleryController	35KB	Photo gallery with album management
VideoGalleryController	24KB	Video gallery
MediaUploadsController	18KB	Generic media upload
FooterSliderItemsController	12KB	Footer slider items
QrCodesController	23KB	QR code generation

3.9 Public Interaction Management

Controller	Size	Purpose
PublicConsultantsController	48KB	Public consultation management
PublicConsultantCommentsController	14KB	Comment moderation
PublicConsultantTrackingDetailsController	9KB	Tracking details
GrievanceController	14KB	Grievance management
ContactUsDetailsController	17KB	Contact form submissions
CommentsController	12KB	Comment management
AeComplaintSuggestionController	15KB	AE complaints/suggestions

3.10 RTI & Publications

Controller	Size	Purpose
RtiApplicationsController	24KB	RTI applications
RtiDisclosureController	23KB	RTI disclosures
RtiJournalController	13KB	RTI journal

JournalController	18KB	Journal management
JournalVolumesController	18KB	Volume management
JournalIssuesController	20KB	Issue management
JournalArticlesController	22KB	Article management
ResearchPaperController	23KB	Research papers
CorePublishedPapersController	18KB	Published papers

3.11 All Remaining Admin Controllers

Controller	Size	Purpose
AdministrativeInformationController	26KB	Administrative info
AgOtherReportsController	22KB	AG other reports
ApplicantProformaController	26KB	Applicant proforma
BlueEconomyCenterController	19KB	Blue economy content
BoardCommitteesController	22KB	Board committees
BookingHostelRoomsController	13KB	Hostel room booking
BudgetController	23KB	Budget documents
CollaborationsController	20KB	Collaborations
CombinedAccountsController	23KB	Combined accounts
CountriesController	10KB	Countries list
DistrictsController	7KB	Districts list
DrawingDisbOfficersController	16KB	Drawing/disbursing officers
EventsController	21KB	Events
FormerCagController	12KB	Former CAGs
FormsController	21KB	Dynamic forms builder
FrmlImportController	21KB	Form data import
FieldsController	10KB	Form fields
GeneralCategoriesController	18KB	Generic categories
GuidelinesController	24KB	Guidelines
GuidanceNotesPracticeGuidesController	25KB	Practice guides
laadKmsContentController	38KB	KMS content
InHouseSystemsController	20KB	In-house systems
KmsCategoriesController	13KB	KMS categories
LibraryController	10KB	Library
ManualsController	15KB	Manuals
NoticeController	19KB	Notices
NotificationController	19KB	Notifications
OfficeMannualController	20KB	Office manuals
OrgChargeMasterController	14KB	Org charge master
OutstandingTreasuryInspectionReportController	19KB	Treasury reports
PerformanceActivityReportController	24KB	Performance reports
PensionController	15KB	Pension management
PolicyAndGuidelinesController	20KB	Policies
RajbhashaCadreController	22KB	Hindi language cadre
RecruitmentNoticesController	27KB	Recruitment
RecruitmentRulesController	26KB	Recruitment rules
ResourcePersonsController	21KB	Resource persons
StaffAssociationsJcmController	12KB	Staff associations
StudyReportsController	15KB	Study reports

TendersController	27KB	Tenders
TrainingController	10KB	Training
TrainingCalendarController	22KB	Training calendar
TrainingEventsController	22KB	Training events
TreasuryOfficersController	23KB	Treasury officers
UserOfficesController	14KB	User offices
WelfareController	22KB	Welfare
AjaxController	17KB	Admin AJAX endpoints
ErrorController	2KB	Admin error handling
functions.php	240B	Legacy function namespace shim

09 — ORM Models & Entities

The application contains **187 Table models** (`src/Model/Table/`) and **182 Entity classes** (`src/Model/Entity/`). It implements several custom extensions to support **PostgreSQL case-sensitivity rules** and **custom JSON/CSV data types**.

1. ORM Architecture Diagram

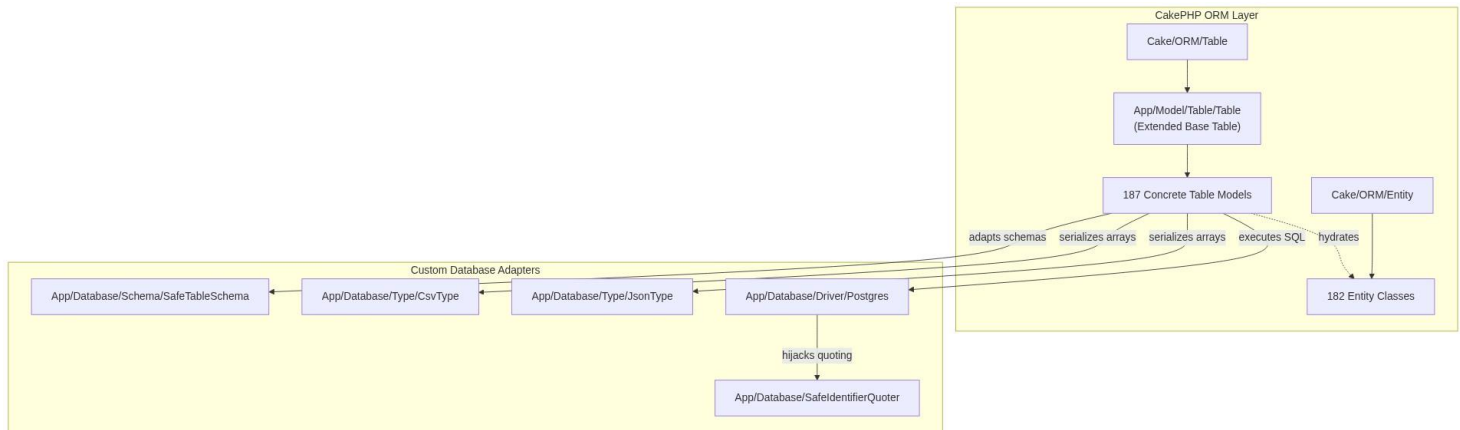


Figure: 09_models_and_entities Diagram 1

2. The Custom Base Class:

[Table.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Model/Table/Table.php)

Every table model in the application extends `App\Model\Table\Table` instead of `Cake\ORM\Table`. This class is a **critical compatibility layer** containing ~600 lines of custom SQL generation logic.

2.1 Why It Exists

The application was originally built on MySQL (where identifiers and aliases are case-insensitive by default) and subsequently migrated to **PostgreSQL** (which forces all unquoted identifiers to lowercase).

If a CakePHP query specifies a CamelCase alias (e.g., `GovernmentTypeData`), Postgres converts it to `governmenttypedata` unless quoted as `"GovernmentTypeData"`. The base `Table` class intercepts all queries and dynamically quotes aliases.

2.2 Key Methods

Method	Operation
<code>initialize()</code>	Wraps the native <code>TableSchema</code> in <code>SafeTableSchema</code>
<code>addCols()</code>	Adds columns to select, intercepting subqueries and <code>CASE WHEN</code> statements to prevent quotation errors
<code>addJoin()</code>	Intercepts SQL joins, dynamically running

	<code>quoteIdentifiersInCondition()</code> on join clauses
<code>quoteIdentifiersInCondition()</code>	Regex-quotes CamelCase identifiers (e.g. <code>TableName.column_name</code> → <code>"TableName".column_name</code>)
<code>beforeFind()</code>	Fires custom query adjustments before database execution

3. PostgreSQL Database Drivers

3.1 Custom Driver:

`[Postgres.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Database/Driver/Postgres.php)`

Extends `Cake\Database\Driver\Postgres` and overrides the `quoter()` method to return `SafeIdentifierQuoter` instead of the default CakePHP quoter.

3.2 Safe Identifier Quoter:

`[SafeIdentifierQuoter.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Database/SafeIdentifierQuoter.php)`

Overrides `quoteIdentifier()`. If a string matches the SQL expression pattern (contains SQL keywords like `CASE`, `CAST`, `EXTRACT`, `ILIKE`, `TO_CHAR`, or standard operators like `>`, `<`, `=`), it returns the string **unmodified**. Simple fields are quoted as normal. This prevents SQL errors where Postgres tries to parse a whole formula as a column name.

4. Custom Database Types

These are registered in `config/bootstrap.php` and mapped in Table schemas.

4.1 CsvType

`[CsvType.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Database/Type/CsvType.php)`

- **Purpose:** Automatically serializes PHP arrays into comma-separated strings for database insertion, and explodes comma-separated strings back into PHP arrays on retrieval.
- **Methods:** `toDatabase()`, `toPHP()`, `marshal()`

4.2 JsonType

`[JsonType.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Database/Type/JsonType.php)`

- **Purpose:** Handles fields storing JSON data, decoding JSON strings into PHP associative arrays automatically.

5. Model Validation

Validations are defined in `validationDefault(Validator $validator)` methods.

5.1 Whitelist File Validation

File fields use a custom validator provider registered on-the-fly:

```
$validator->setProvider('fileValidator',  
    \App\Model\Validation\FileValidatorRules::class);
```

This validates files at the Model level, checking file signatures, sizes, and file extension white-lists.

5.2 Regular Expression Whitelisting

Fields containing user names, titles, and comments are whitelisted against security patterns:

- `REGEX_FOR_NAME` (configured in constants) — Blocks special characters in names
- `REGEX_FOR_TEXT` — Validates alphanumeric structures

6. Entity Behaviors & Virtual Fields

Entity classes (`src/Model/Entity/`) represent database records. They contain standard mutators (setters/getters) and virtual properties:

6.1 Password Auto-Hashing

The `User` entity hashes passwords automatically on modification using bcrypt:

```
protected function _setPassword($password)  
{  
    if (strlen($password) > 0) {  
        return (new DefaultPasswordHasher())->hash($password);  
    }  
    return $this->_fields['password'] ?? '';  
}
```

6.2 Virtual Properties

Entities define virtual fields using the prefix pattern `_get{FieldName}`:

- ``full_name``: Combines `first_name`, `middle_name`, and `last_name` with spaces
- ``petname``: Generates initials (e.g. "John Doe" → "JD") for avatars
- ``translation_title``: Dynamically returns the translated title based on language parameters

10 — View Helpers & Controller Components

The application leverages custom View Helpers and Controller Components to keep controllers thin and handle complex operations like dynamic form rendering, menu security, file attachments, and captchas.

1. Custom View Helpers (8 Files)

View Helpers live in `src/View/Helper/` and are called from PHP view templates.

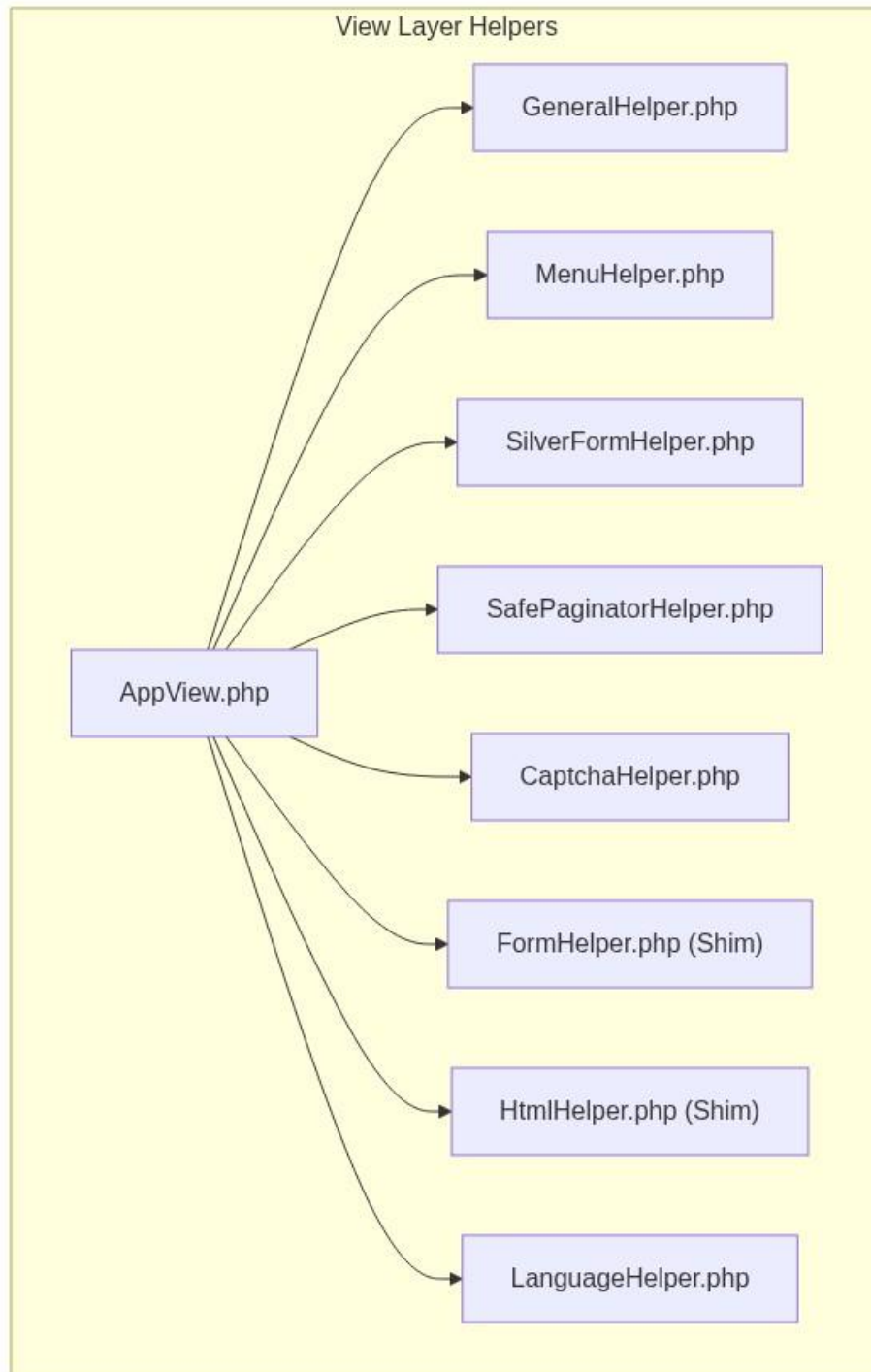


Figure: 10_views_helpers_components Diagram 1

1.1 AppView Configuration

Defined in [AppView.php](#).

- Configures **SafePaginator** as the default pagination renderer
- Adds a **URL Filter** that auto-injects the active **website_id** query parameter into all links generated in templates if the user is authenticated in a non-admin role. This preserves sub-site context during navigation.

1.2 MenuHelper

([MenuHelper.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/View/Helper/MenuHelper.php))

- **File Size:** 37KB (822 lines)
- **Role:** Generates multi-level navigation menus dynamically based on sub-site and user roles.
- **Key Methods:**
- `get($menu_region_slug, $options)` — Fetches the menu tree for a region, evaluating both public permissions (**public**, **authorized**, **unauthorized**) and role-based permissions (**private_access** matches logged-in user role).
- `getMenuGroups()` — Fetches grouped menus for admin columns.

1.3 SilverFormHelper

([SilverFormHelper.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/View/Helper/SilverFormHelper.php))

- **File Size:** 70KB (1515 lines)
- **Role:** The engine for **dynamic public forms**. It generates forms (inputs, textareas, file uploads, drop-downs) based on configurations in the **forms** and **fields** database tables.
- **Key Methods:**
- `get($form_id)` — Fetches the form fields and generates HTML elements dynamically.
- `save($form_id, $submit_id, $data)` — Saves form submissions into dynamic sub-tables named **silver_form_{form_id}**.
- `validate($form_id)` — Evaluates dynamically configured validation rules (required, formats).

1.4 SafePaginatorHelper

([SafePaginatorHelper.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/View/Helper/SafePaginatorHelper.php))

- **Role:** A security-hardened wrapper for CakePHP's native paginator helper. It sanitizes pagination parameters to prevent SQL injection or parameter manipulation.

1.5 CaptchaHelper

([CaptchaHelper.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/View/Helper/CaptchaHelper.php))

- **Role:** Renders captcha images on public forms (e.g. login, feedback). Generates a random mathematical equation or string and outputs the image element linked to the captcha controller.

2. Controller Components (8 Files)

Controller Components live in `src/Controller/Component/` and share logic between controllers.

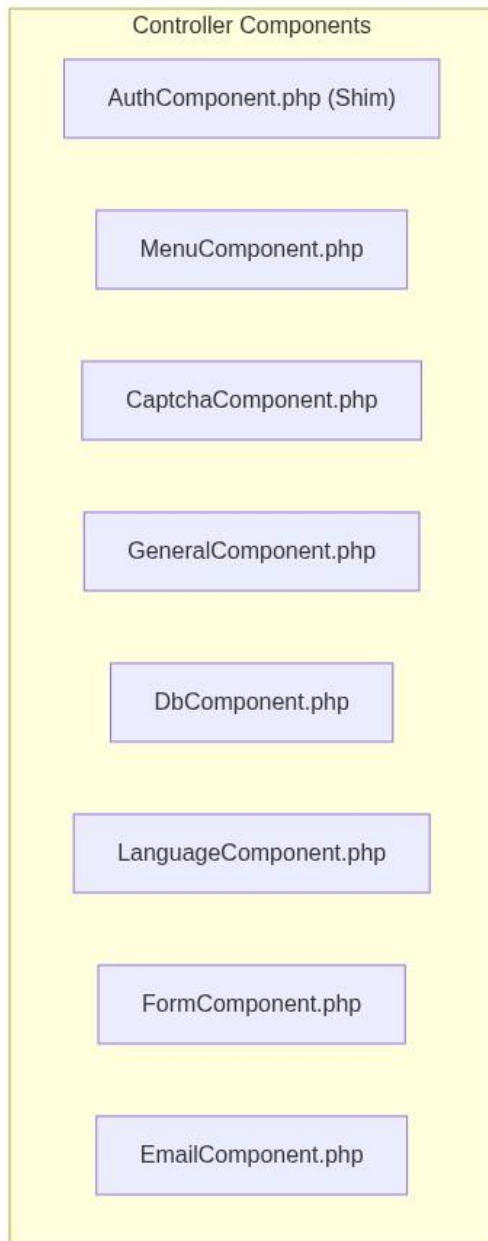


Figure: 10_views_helpers_components Diagram 2

2.1 AuthComponent

- **Details:** Crucial shim mapping CakePHP 3 `AuthComponent` API calls to CakePHP 5 `AuthenticationMiddleware`. Detailed in [04_authentication_authorization.md](#).

2.2 MenuComponent

([MenuComponent.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Controller/Component/MenuComponent.php))

- **Role:** Exposes menu permission checks to controllers.

- **Methods:** `hasAccess($menu_region_slug, $options)` redirects users if they lack permission to view a page mapped to a menu item.

2.3 CaptchaComponent

([CaptchaComponent.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Controller/Component/CaptchaComponent.php))

- **File Size:** 17KB (486 lines)
- **Role:** Code generator for captchas. Uses PHP GD library to draw characters onto an image with random background lines, distortion, and font rotations (using fonts located in `src/Lib/Fonts/`). Stores the true value in session for validation.

2.4 DbComponent

([DbComponent.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Controller/Component/DbComponent.php))

- **Role:** Abstraction layer for raw SQL execution.
- **Methods:** `execute()`, `update()`, `insert()`, `createQuery()` wrap standard CakePHP connection manager executions for legacy code.

2.5 FormComponent

([FormComponent.php](file:///c:/Users/yokes/OneDrive/Desktop/new%20Analysis/CAG_Website/src/Controller/Component/FormComponent.php))

- **Role:** Creates and maintains dynamic tables for the `SilverForm` engine.
- **Key Method:** `createTable($table_name)` issues a `CREATE TABLE` query for submissions and a corresponding clone table for audit trails.

11 — End-to-End Workflows

This document details the visual logic flow for critical system tasks.

1. Public Content Delivery & Multi-Site Flow

When a user visits any public page on the main site or a sub-site:

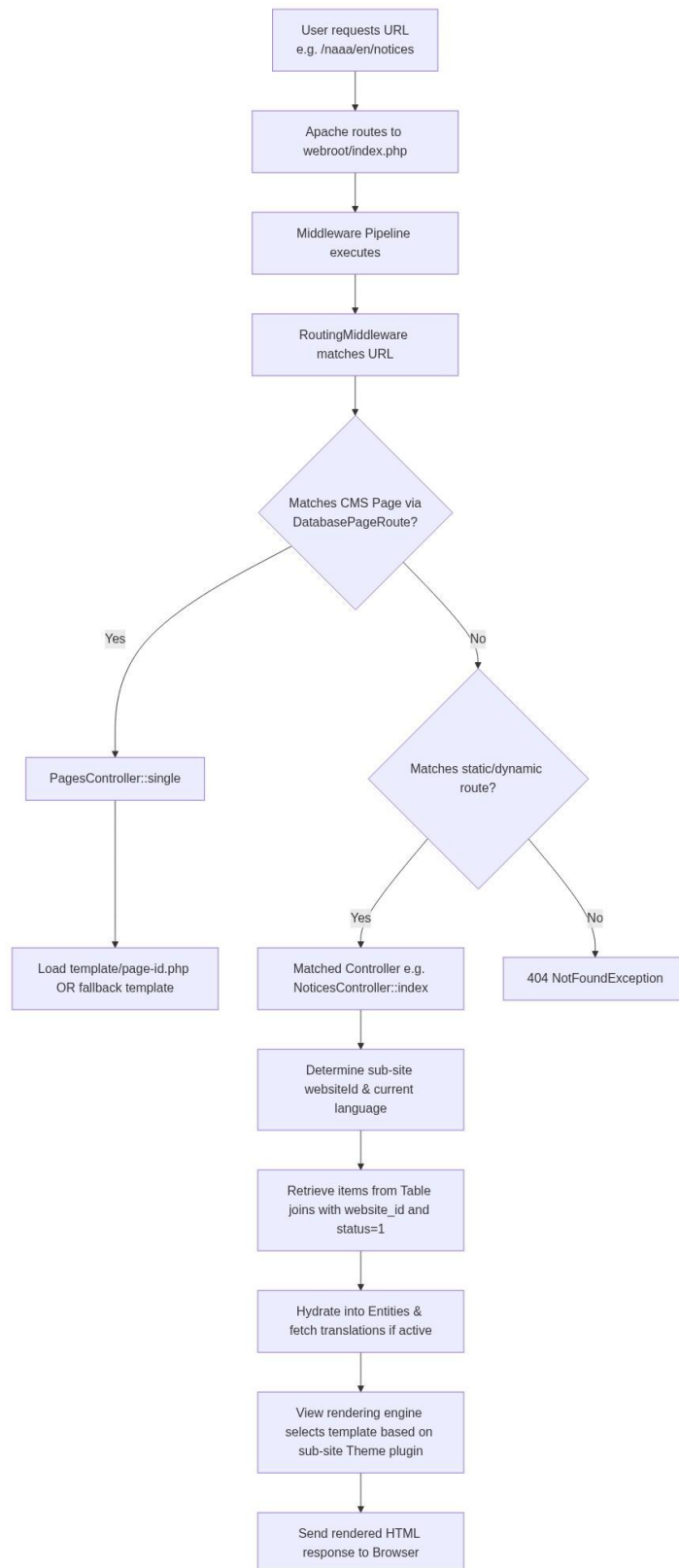


Figure: 11_end_to_end_workflow Diagram 1

2. Dynamic Form Submission Flow (SilverForm Engine)

How user-submitted data on public pages (such as registration forms, internship applications) is created, processed, validated, and stored dynamically:

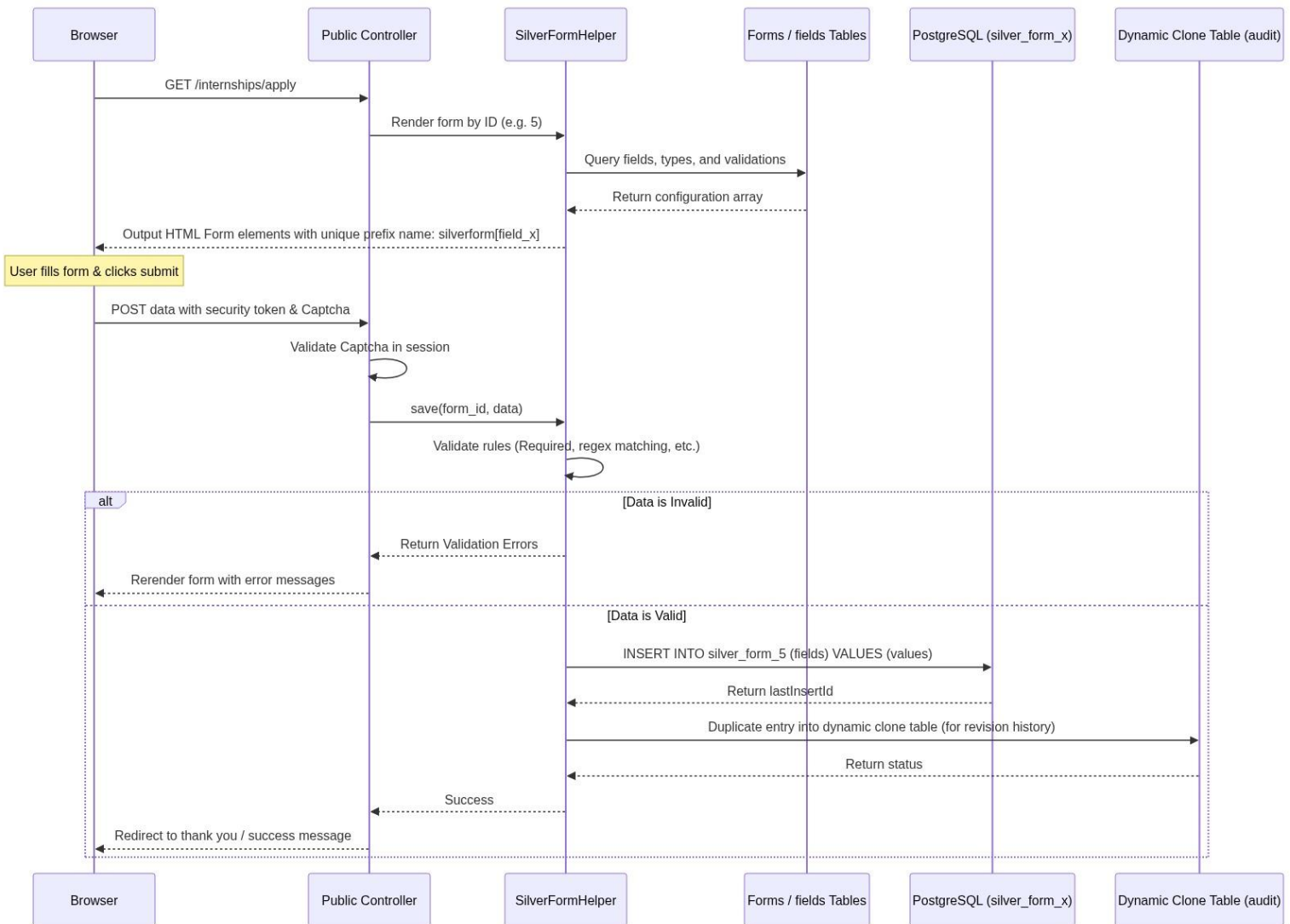


Figure: 11_end_to_end_workflow Diagram 2

3. Secure File Upload Lifecycle

To prevent remote code execution (RCE) on the server, all file uploads undergo a multi-layered security filter before they are written to the filesystem.

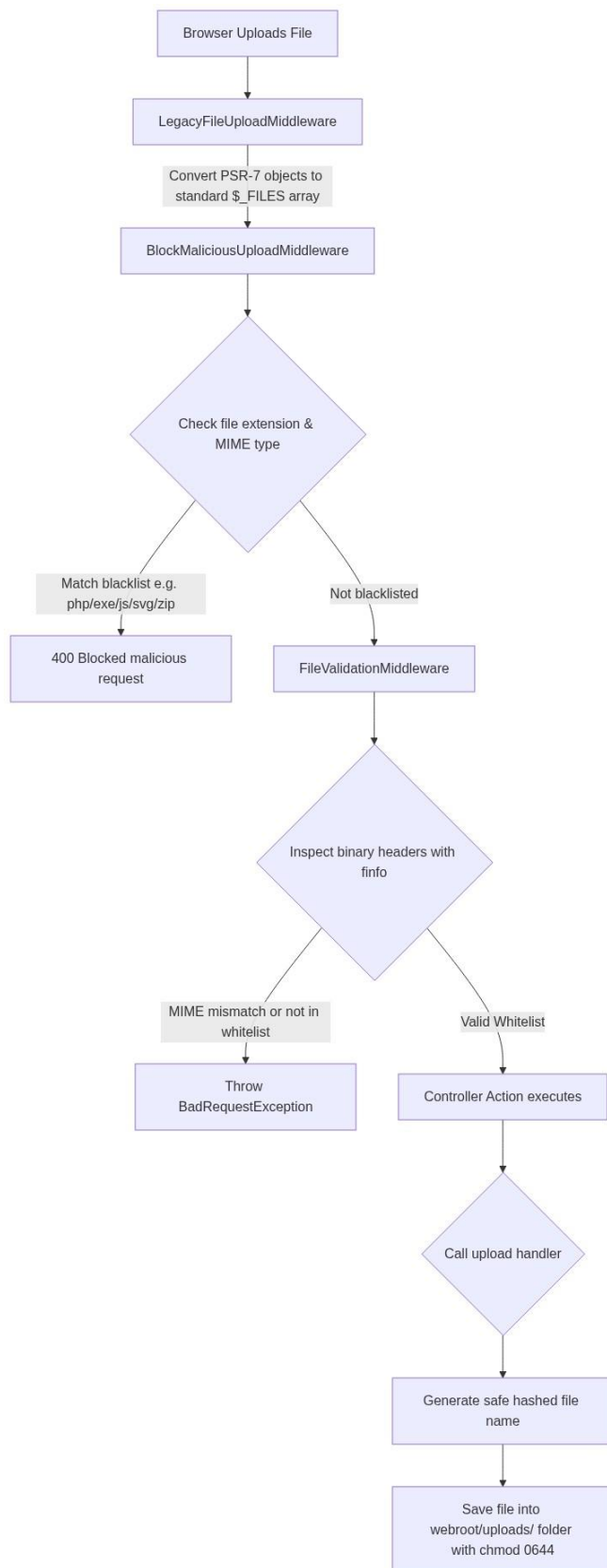


Figure: 11_end_to_end_workflow Diagram 3

4. Admin Content Publishing Workflow

How administrative users create and publish updates (e.g., Audit Reports, Circulars):

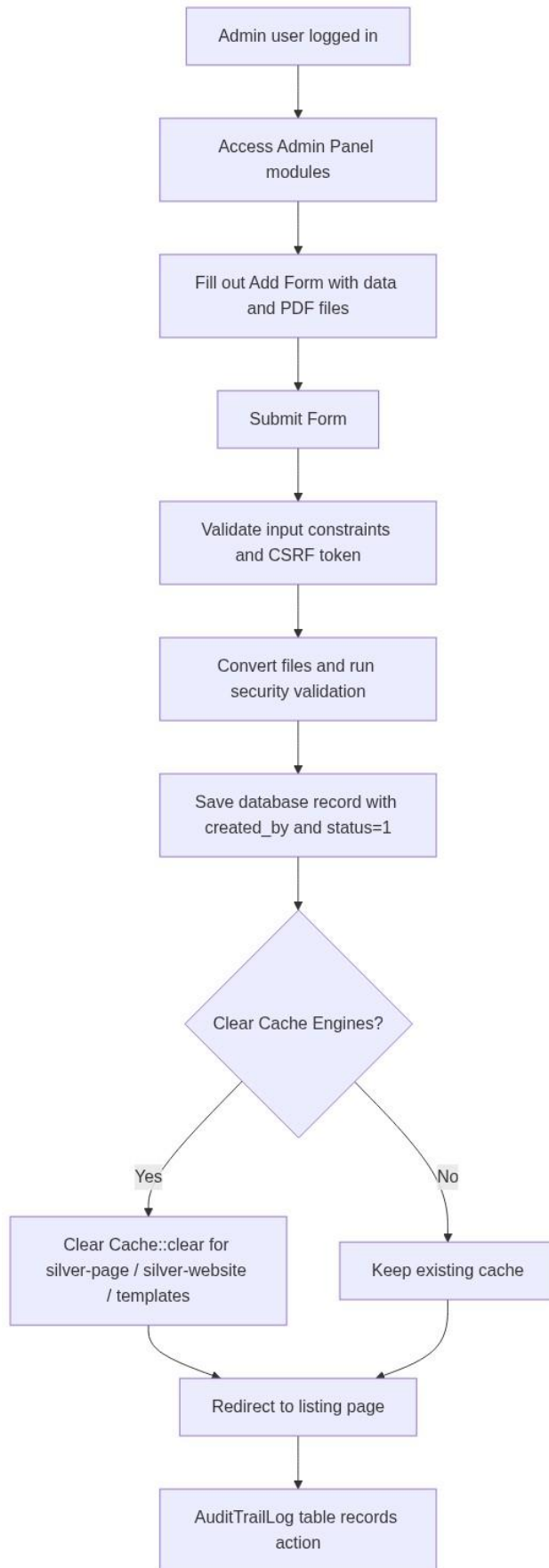


Figure: 11_end_to_end_workflow Diagram 4

12 — APIs & Integrations

The system exposes interfaces for machine-to-machine integrations, JSON/XML APIs, and Single Sign-On (SSO) authentication.

1. API Route Prefixes

The application exposes two prefixes for external integrations:

Route Prefix	Controller Path	Response Formats	Purpose
/api	src/Controller/Api/	JSON, XML	Internal JSON API endpoints
/webapi	src/Controller/Webapi/	JSON	Public consumption endpoints

2. Dynamic Format Negotiation

API controllers enable JSON and XML output parsing natively in CakePHP 5:

- Routes in the /api prefix register:

```
$routes->setExtensions(['json', 'xml']);
```

- Controllers use the `RequestHandler` mechanism (adapted to CakePHP 5) to automatically serialize array data assigned to view variables. When a request includes the `.json` or `.xml` extension (e.g. `/api/notices.json`), the framework skips rendering PHP templates and outputs the parsed content directly.

3. Parichay SSO Integration

File: [ParichaySsoController.php](#)

The application integrates with **Parichay SSO**, a centralized authentication system managed by the National Informatics Centre (NIC), Government of India. It enables government officials to log into the admin panel securely without using local passwords.

3.1 Authentication Workflow

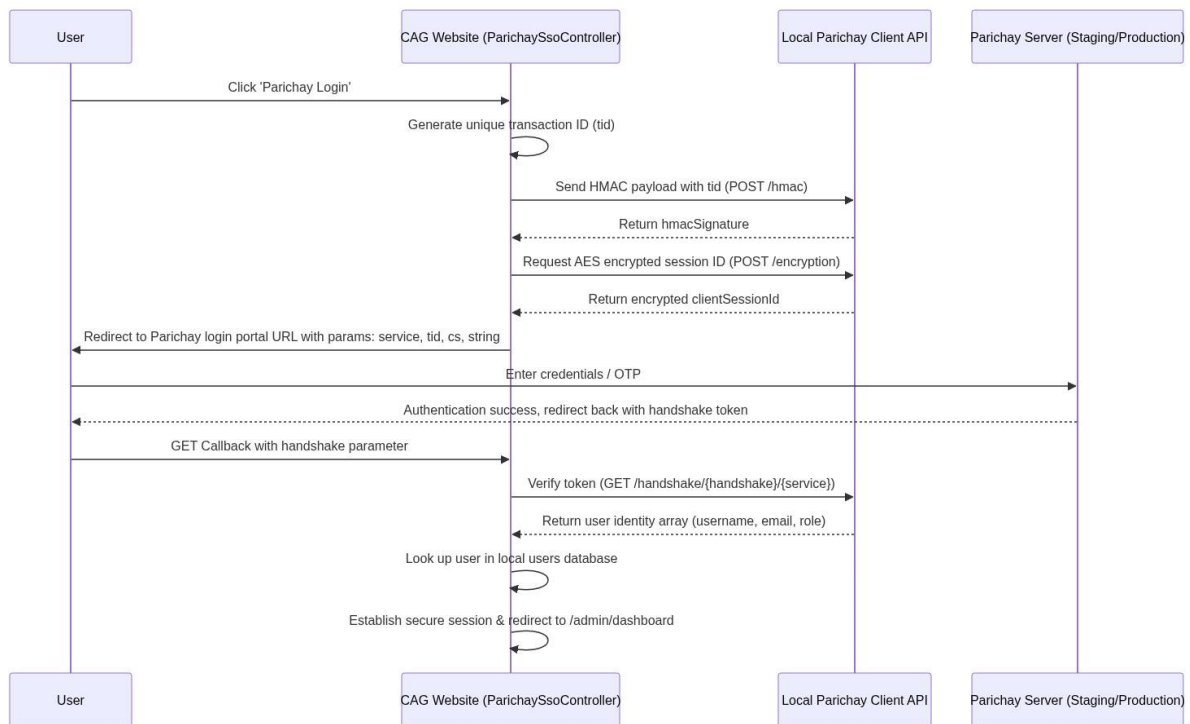


Figure: 12_api_and_integrations Diagram 1

3.2 Key Endpoints & Parameters

- **HMAC Generation URL:** `PARICHAY_CLIENT_URL + /hmac`
- **Encryption URL:** `PARICHAY_CLIENT_URL + /encryption`
- **Handshake Verification URL:** `PARICHAY_CLIENT_URL + /handshake/{handshakeToken}/{serviceName}`
- **Session Check:** `isTokenValid` API endpoint validates active SSO tokens periodically

4. LDAP Authentication

The system supports **LDAP (Lightweight Directory Access Protocol)** authentication for restricted content (such as internal KMS documents).

- **Control Table:** The `circular_categories` table contains the column `is_ldap_required`
- **Controller Action:** In `CircularsController::checkAccess()`:
- If a circular category has `is_ldap_required = 1`, the controller prompts the user to enter their official NIC email credentials
- The request binds to the configured LDAP server (using the PHP `ldap` extension)
- If successful, the circular PDF is served securely; otherwise, access is blocked

5. Security & CORS for Integrations

1. **Host Header Inspection:** The `HostHeaderMiddleware` ensures incoming API calls use the whitelisted server domain name, protecting against Host Header attacks.
2. **Referer Checking:** Post requests are restricted to matching origin hosts in [Admin/AppController::initialize\(\)](#).
3. **Query Sanitization:** The `SanitizeRequestMiddleware` validates all query parameters for API calls, blocking malicious injection payloads.